

卫星仿真平台工程级模型验证测试细则

/ satellite-simulation-platform 0.5.3 / 0.5.3

1. 文档范围与结论

本文档面向 工程级模型验证测试体系。测试范围覆盖环境、静态质量、标定证据、能力注册、部件级模型、分系统、整星代表场景、Agent 规划、Golden Set 和发布收口。

本细则以机器可执行矩阵 configs/verification/engineering_verification_matrix.json 为权威来源；每个测试项均给出可直接执行的命令、判读参数和通过/失败规则。

边界：验证当前已注册且已实现的公开 Runner、事件/退化机制和 Agent 链路，不声称覆盖未建模物理、真实标定或飞行认证。

统计项	数值
矩阵版本	.engineering-verification-matrix.v1
测试项总数	437
部件数量	24
分系统数量	6
部件级测试项	392
分系统测试项	12
环境/静态/模型/标定/Agent/发布测试项	33

2. 通用执行方式

推荐从工程根目录执行以下命令。Linux/WSL：

```
export PYTHONPATH=./src
python scripts/run_engineering_verification_matrix.py
--statspython scripts/check_engineering_verification_matrix.py
x.pypython scripts/run_engineering_verification_matrix.py --level component --execute-all --output-dir reports/既有工程版本_c
omponent_matrixpython scripts/run_engineering_verification_matrix.py --level subsystem --execute-all --output-dir reports/既有工程版本
_subsystem_matrix
```

单项测试命令格式：

```
PYTHONPATH=./src python scripts/run_engineering_verification_matrix.py --test-id <TEST_ID> --output reports//<TEST_ID>.json
```

3. 判读通用规则

结果	含义	研判规则
PASS	测试项满足判读参数和通过规则	可进入下一层测试；仍需保留 JSON 证据。
FAIL	命令或自动判读失败	不得以人工解释替代；需定位 reason、stderr、Run Bundle 或 QoI。
INCONCLUSIVE	执行完成但证据不足	不能提升为 PASS，需补 QoI、数据或模型。
NOT_EVALUATED	执行失败或取消	不能评价任务目标。

禁止规则：不得删除 QoI 后判 PASS；不得放宽门限后判 PASS；不得将 proxy 声明为 native；不得把 delivery 成功等同于 effect 成功；不得把 mission FAIL 自动修复为 PASS。

4. 测试矩阵分布

Level	数量
agent	6
calibration	4
component	392
environment	2
model_contract	7
release	2
static_quality	6
subsystem	12
whole_spacecraft_or_agent_release	6
Scenario Type	数量
all_modes	6
batch	24
combined_default_set	24
compileall	1
degradation	48
degradation_default_set	24
determinism	24
duplicate_definitions	1
duration_boundary	48
exception_policy	1
export_script	1
fault	48
fault_default_set	24
golden_repeat5	1
invalid_ground_calibrated_rejection	1
layer_schema	1
native_capability_audit	1
negative	24
nominal	24

Level	数量
normal_working	80
parameter_provenance	1
parse_template	1
plan_whole	1
public_exports	1
public_type_hints	1
pytest	1
readiness_check	1
readiness_report	1
release_check	1
repeat_stability	6
representative_run	6
resolve_component	1
ruff_blockers	1
runtime_injection	1
schema	1
strict_doctor	1
subsystem_builders	1
subsystem_runners	1
validate_component	1
version	1
whole_spacecraft_native	1

5. 部件级测试覆盖

部件	测试项数量	覆盖场景类型
antenna	17	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1, normal_working:4
battery	17	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1, normal_working:4
cmg	17	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1, normal_working:4
data_queue	17	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1, normal_working:4
fuel_tank	17	batch:1, combined_default_set:1, degradation:2,

部件	测试项数量	覆盖场景类型
		degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1, normal_working:4
ground_station	17	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1, normal_working:4
heater	13	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1
imu	17	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1, normal_working:4
link_budget	17	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1, normal_working:4
magnetometer	17	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1, normal_working:4
mtb	17	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1, normal_working:4
onboard_storage	17	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1, normal_working:4
payload	17	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1, normal_working:4
payload_sensor	17	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1, normal_working:4
pdu	17	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1, normal_working:4
power_sink	17	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1,

部件	测试项数量	覆盖场景类型
		duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1, normal_working:4
radiator	13	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1
reaction_wheel	13	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1
solar_panel	13	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1
star_tracker	17	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1, normal_working:4
sun_sensor	17	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1, normal_working:4
thermal_node	17	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1, normal_working:4
thruster	17	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1, normal_working:4
transmitter	17	batch:1, combined_default_set:1, degradation:2, degradation_default_set:1, determinism:1, duration_boundary:2, fault:2, fault_default_set:1, negative:1, nominal:1, normal_working:4

6. 完整测试项明细（437 项）

下表列出所有测试项。命令均可直接从工程根目录执行。判读参数采用 JSON 摘要；完整版本见 configs/verification/engineering_verification_matrix.json。

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
CMP-NOM-ANTENNA-NOMINAL	component	antenna	nominal / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-	{"finite_numeric_required":true,"error_key_forbidden":true}	run_nominal_case returns structured data without error and all numeric outputs are finite.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				NOM-ANTENNA-NOMINAL --output reports//CMP-NOM-ANTENNA-NOMINAL.json		
CMP-WRK-ANTENNA-NOMINAL	component	antenna	normal_working / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-ANTENNA-NOMINAL --output reports//CMP-WRK-ANTENNA-NOMINAL.json	{"finite_numeric_required":true,"scenario":"nominal"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-ANTENNA-BORESIGHT-TRACKING	component	antenna	normal_working / boresight_tracking	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-ANTENNA-BORESIGHT-TRACKING --output reports//CMP-WRK-ANTENNA-BORESIGHT-TRACKING.json	{"finite_numeric_required":true,"scenario":"boresight_tracking"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-ANTENNA-SLEW-ACQUISITION	component	antenna	normal_working / slew_acquisition	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-ANTENNA-SLEW-ACQUISITION --output reports//CMP-WRK-ANTENNA-SLEW-ACQUISITION.json	{"finite_numeric_required":true,"scenario":"slew_acquisition"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-ANTENNA-HALF-POWER-EDGE	component	antenna	normal_working / half_power_edge	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-ANTENNA-HALF-POWER-EDGE	{"finite_numeric_required":true,"scenario":"half_power_edge"}	Named normal operating scenario executes and returns finite outputs with no error field.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				--output reports//CMP-WRK- ANTENNA-HALF- POWER-EDGE.json		
CMP-ALL-ANTENNA-RUN-ALL-MODES	component	antenna	batch / run_all_modes	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- ALL-ANTENNA-RUN- ALL-MODES --output reports//CMP-ALL- ANTENNA-RUN-ALL- MODES.json	{"minimum_top_level_modes":3,"fi nite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2- light coverage and finite numeric outputs.
CMP-FLT-ANTENNA-POINTING- LOSS	component	antenna	fault / pointing_loss	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- FLT-ANTENNA- POINTING-LOSS -- output reports//CMP- FLT-ANTENNA- POINTING-LOSS.json	{"active_fault_required":true,"sam ple_window_s":{0,600}}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-FLT-ANTENNA-FEED-OPEN	component	antenna	fault / feed_open	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- FLT-ANTENNA-FEED- OPEN --output reports//CMP-FLT- ANTENNA-FEED- OPEN.json	{"active_fault_required":true,"sam ple_window_s":{0,600}}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-DEG-ANTENNA-SURFACE- CONTAMINATION	component	antenna	degradation / surface_contamination	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- DEG-ANTENNA- SURFACE- CONTAMINATION -- output reports//CMP- DEG-ANTENNA- SURFACE-	{"active_degradation_required":tru e,"sample_window_s":{0,600}}	The single degradation is active after onset and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				CONTAMINATION.json		
CMP-DEG-ANTENNA-POINTING-BIAS-DRIFT	component	antenna	degradation / pointing_bias_drift	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-ANTENNA-POINTING-BIAS-DRIFT --output reports//CMP-DEG-ANTENNA-POINTING-BIAS-DRIFT.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-L2-ANTENNA-FAULT-DEFAULT-SET	component	antenna	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-ANTENNA-FAULT-DEFAULT-SET --output reports//CMP-L2-ANTENNA-FAULT-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.
CMP-L2-ANTENNA-DEGRADATION-DEFAULT-SET	component	antenna	degradation_default_set / degradation_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-ANTENNA-DEGRADATION-DEFAULT-SET --output reports//CMP-L2-ANTENNA-DEGRADATION-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default degradations participate at least once.
CMP-L2-ANTENNA-COMBINED-DEFAULT-SET	component	antenna	combined_default_set / combined_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-ANTENNA-COMBINED-DEFAULT-SET --output reports//CMP-	{"duration_s":600,"finite_numeric_required":true}	At least one fault and one degradation participate in one run.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				L2-ANTENNA-COMBINED-DEFAULT-SET.json		
CMP-BND-ANTENNA-SHORT-DURATION-60S	component	antenna	duration_boundary / short_duration_60s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-ANTENNA-SHORT-DURATION-60S --output reports//CMP-BND-ANTENNA-SHORT-DURATION-60S.json	{"duration_s":60.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-BND-ANTENNA-LONG-DURATION-86400S	component	antenna	duration_boundary / long_duration_86400s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-ANTENNA-LONG-DURATION-86400S --output reports//CMP-BND-ANTENNA-LONG-DURATION-86400S.json	{"duration_s":86400.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-NEG-ANTENNA-UNSUPPORTED-WORKING-SCENARIO	component	antenna	negative / unsupported_working_scenario	python scripts/run_engineering_verification_matrix.py --test-id CMP-NEG-ANTENNA-UNSUPPORTED-WORKING-SCENARIO --output reports//CMP-NEG-ANTENNA-UNSUPPORTED-WORKING-SCENARIO.json	{"expected_exception":"ValueError"}	Unsupported working scenario is rejected by the component runner.
CMP-DET-ANTENNA-REPEAT-STABILITY	component	antenna	determinism / repeat_stability	python scripts/run_engineeri	{"repeat_count":2,"semantic_hash_equal":true}	Two same-input runs produce the same semantic hash.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				ng_verification_matrix.py --test-id CMP-DET-ANTENNA-REPEAT-STABILITY --output reports//CMP-DET-ANTENNA-REPEAT-STABILITY.json		
CMP-NOM-BATTERY-NOMINAL	component	battery	nominal / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-NOM-BATTERY-NOMINAL --output reports//CMP-NOM-BATTERY-NOMINAL.json	{"finite_numeric_required":true,"error_key_forbidden":true}	run_nominal_case returns structured data without error and all numeric outputs are finite.
CMP-WRK-BATTERY-NOMINAL	component	battery	normal_working / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-BATTERY-NOMINAL --output reports//CMP-WRK-BATTERY-NOMINAL.json	{"finite_numeric_required":true,"scenario":"nominal"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-BATTERY-CHARGE-WINDOW	component	battery	normal_working / charge_window	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-BATTERY-CHARGE-WINDOW --output reports//CMP-WRK-BATTERY-CHARGE-WINDOW.json	{"finite_numeric_required":true,"scenario":"charge_window"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-BATTERY-ECLIPSE-DISCHARGE	component	battery	normal_working / eclipse_discharge	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-BATTERY-	{"finite_numeric_required":true,"scenario":"eclipse_discharge"}	Named normal operating scenario executes and returns finite outputs with no error field.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				ECLIPSE-DISCHARGE --output reports//CMP-WRK-BATTERY-ECLIPSE-DISCHARGE.json		
CMP-WRK-BATTERY-ORBIT-CYCLE	component	battery	normal_working / orbit_cycle	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-BATTERY-ORBIT-CYCLE --output reports//CMP-WRK-BATTERY-ORBIT-CYCLE.json	{"finite_numeric_required":true,"scenario":"orbit_cycle"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-ALL-BATTERY-RUN-ALL-MODES	component	battery	batch / run_all_modes	python scripts/run_engineering_verification_matrix.py --test-id CMP-ALL-BATTERY-RUN-ALL-MODES --output reports//CMP-ALL-BATTERY-RUN-ALL-MODES.json	{"minimum_top_level_modes":3,"finite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2-light coverage and finite numeric outputs.
CMP-FLT-BATTERY-OPEN-CIRCUIT	component	battery	fault / open_circuit	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-BATTERY-OPEN-CIRCUIT --output reports//CMP-FLT-BATTERY-OPEN-CIRCUIT.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-FLT-BATTERY-INTERNAL-SHORT	component	battery	fault / internal_short	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-BATTERY-INTERNAL-SHORT --output reports//CMP-FLT-BATTERY-INTERNAL-	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				SHORT.json		
CMP-DEG-BATTERY-CYCLE-CAPACITY-FADE	component	battery	degradation / cycle_capacity_fade	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-BATTERY-CYCLE-CAPACITY-FADE --output reports//CMP-DEG-BATTERY-CYCLE-CAPACITY-FADE.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-DEG-BATTERY-RESISTANCE-GROWTH	component	battery	degradation / resistance_growth	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-BATTERY-RESISTANCE-GROWTH --output reports//CMP-DEG-BATTERY-RESISTANCE-GROWTH.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-L2-BATTERY-FAULT-DEFAULT-SET	component	battery	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-BATTERY-FAULT-DEFAULT-SET --output reports//CMP-L2-BATTERY-FAULT-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.
CMP-L2-BATTERY-DEGRADATION-DEFAULT-SET	component	battery	degradation_default_set / degradation_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-BATTERY-DEGRADATION-DEFAULT-SET --output reports//CMP-L2-BATTERY-DEGRADATION-	{"duration_s":600,"finite_numeric_required":true}	All default degradations participate at least once.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				DEFAULT-SET.json		
CMP-L2-BATTERY-COMBINED-DEFAULT-SET	component	battery	combined_default_set / combined_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-BATTERY-COMBINED-DEFAULT-SET --output reports//CMP-L2-BATTERY-COMBINED-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	At least one fault and one degradation participate in one run.
CMP-BND-BATTERY-SHORT-DURATION-60S	component	battery	duration_boundary / short_duration_60s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-BATTERY-SHORT-DURATION-60S --output reports//CMP-BND-BATTERY-SHORT-DURATION-60S.json	{"duration_s":60.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-BND-BATTERY-LONG-DURATION-86400S	component	battery	duration_boundary / long_duration_86400s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-BATTERY-LONG-DURATION-86400S --output reports//CMP-BND-BATTERY-LONG-DURATION-86400S.json	{"duration_s":86400.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-NEG-BATTERY-UNSUPPORTED-WORKING-SCENARIO	component	battery	negative / unsupported_working_scenario	python scripts/run_engineering_verification_matrix.py --test-id CMP-NEG-BATTERY-UNSUPPORTED-	{"expected_exception":"ValueError"}	Unsupported working scenario is rejected by the component runner.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				WORKING- SCENARIO --output reports//CMP-NEG- BATTERY- UNSUPPORTED- WORKING- SCENARIO.json		
CMP-DET-BATTERY-REPEAT-STABILITY	component	battery	determinism / repeat_stability	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- DET-BATTERY- REPEAT-STABILITY -- output reports//CMP- DET-BATTERY- REPEAT- STABILITY.json	{"repeat_count":2,"semantic_hash_equal":true}	Two same-input runs produce the same semantic hash.
CMP-NOM-CMG-NOMINAL	component	cmg	nominal / nominal	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- NOM-CMG-NOMINAL --output reports//CMP-NOM- CMG-NOMINAL.json	{"finite_numeric_required":true,"error_key_forbidden":true}	run_nominal_case returns structured data without error and all numeric outputs are finite.
CMP-WRK-CMG-NOMINAL	component	cmg	normal_working / nominal	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- WRK-CMG-NOMINAL --output reports//CMP-WRK- CMG-NOMINAL.json	{"finite_numeric_required":true,"scenario":"nominal"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-CMG-SMALL-SLEW	component	cmg	normal_working / small_slew	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- WRK-CMG-SMALL- SLEW --output reports//CMP-WRK- CMG-SMALL-	{"finite_numeric_required":true,"scenario":"small_slew"}	Named normal operating scenario executes and returns finite outputs with no error field.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				SLEW.json		
CMP-WRK-CMG-SLEW-REVERSAL	component	cmg	normal_working / slew_reversal	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-CMG-SLEW-REVERSAL --output reports//CMP-WRK-CMG-SLEW-REVERSAL.json	{"finite_numeric_required":true,"scenario":"slew_reversal"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-CMG-WHEEL-SPEED-TRIM	component	cmg	normal_working / wheel_speed_trim	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-CMG-WHEEL-SPEED-TRIM --output reports//CMP-WRK-CMG-WHEEL-SPEED-TRIM.json	{"finite_numeric_required":true,"scenario":"wheel_speed_trim"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-ALL-CMG-RUN-ALL-MODES	component	cmg	batch / run_all_modes	python scripts/run_engineering_verification_matrix.py --test-id CMP-ALL-CMG-RUN-ALL-MODES --output reports//CMP-ALL-CMG-RUN-ALL-MODES.json	{"minimum_top_level_modes":3,"finite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2-light coverage and finite numeric outputs.
CMP-FLT-CMG-GIMBAL-STUCK	component	cmg	fault / gimbal_stuck	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-CMG-GIMBAL-STUCK --output reports//CMP-FLT-CMG-GIMBAL-STUCK.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-FLT-CMG-WHEEL-DRIVE-FAULT	component	cmg	fault / wheel_drive_fault	python scripts/run_engineering_verification_matrix.py --test-id CMP-	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				FLT-CMG-WHEEL-DRIVE-FAULT --output reports//CMP-FLT-CMG-WHEEL-DRIVE-FAULT.json		
CMP-DEG-CMG-GIMBAL-FRICTION-GROWTH	component	cmg	degradation / gimbal_friction_growth	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-CMG-GIMBAL-FRICTION-GROWTH --output reports//CMP-DEG-CMG-GIMBAL-FRICTION-GROWTH.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-DEG-CMG-MOMENTUM-LOSS	component	cmg	degradation / momentum_loss	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-CMG-MOMENTUM-LOSS --output reports//CMP-DEG-CMG-MOMENTUM-LOSS.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-L2-CMG-FAULT-DEFAULT-SET	component	cmg	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-CMG-FAULT-DEFAULT-SET --output reports//CMP-L2-CMG-FAULT-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.
CMP-L2-CMG-DEGRADATION-DEFAULT-SET	component	cmg	degradation_default_set / degradation_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-CMG-DEGRADATION-DEFAULT-SET --	{"duration_s":600,"finite_numeric_required":true}	All default degradations participate at least once.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				output reports//CMP-L2-CMG-DEGRADATION-DEFAULT-SET.json		
CMP-L2-CMG-COMBINED-DEFAULT-SET	component	cmg	combined_default_set / combined_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-CMG-COMBINED-DEFAULT-SET --output reports//CMP-L2-CMG-COMBINED-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	At least one fault and one degradation participate in one run.
CMP-BND-CMG-SHORT-DURATION-60S	component	cmg	duration_boundary / short_duration_60s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-CMG-SHORT-DURATION-60S --output reports//CMP-BND-CMG-SHORT-DURATION-60S.json	{"duration_s":60.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-BND-CMG-LONG-DURATION-86400S	component	cmg	duration_boundary / long_duration_86400s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-CMG-LONG-DURATION-86400S --output reports//CMP-BND-CMG-LONG-DURATION-86400S.json	{"duration_s":86400.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-NEG-CMG-UNSUPPORTED-WORKING-SCENARIO	component	cmg	negative / unsupported_working_scenario	python scripts/run_engineering_verification_matrix.py --test-id CMP-NEG-CMG-UNSUPPORTED-WORKING-SCENARIO --output reports//CMP-NEG-	{"expected_exception":"ValueError"}	Unsupported working scenario is rejected by the component runner.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				CMG-UNSUPPORTED-WORKING-SCENARIO.json		
CMP-DET-CMG-REPEAT-STABILITY	component	cmg	determinism / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id CMP-DET-CMG-REPEAT-STABILITY --output reports//CMP-DET-CMG-REPEAT-STABILITY.json	{"repeat_count":2,"semantic_hash_equal":true}	Two same-input runs produce the same semantic hash.
CMP-NOM-DATA-QUEUE-NOMINAL	component	data_queue	nominal / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-NOM-DATA-QUEUE-NOMINAL --output reports//CMP-NOM-DATA-QUEUE-NOMINAL.json	{"finite_numeric_required":true,"error_key_forbidden":true}	run_nominal_case returns structured data without error and all numeric outputs are finite.
CMP-WRK-DATA-QUEUE-NOMINAL	component	data_queue	normal_working / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-DATA-QUEUE-NOMINAL --output reports//CMP-WRK-DATA-QUEUE-NOMINAL.json	{"finite_numeric_required":true,"scenario":"nominal"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-DATA-QUEUE-IMAGING-ACCUMULATION	component	data_queue	normal_working / imaging_accumulation	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-DATA-QUEUE-IMAGING-ACCUMULATION --output reports//CMP-WRK-DATA-QUEUE-IMAGING-ACCUMULATION.jso	{"finite_numeric_required":true,"scenario":"imaging_accumulation"}	Named normal operating scenario executes and returns finite outputs with no error field.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				n		
CMP-WRK-DATA-QUEUE-DOWNLINK-PASS	component	data_queue	normal_working / downlink_pass	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-DATA-QUEUE-DOWNLINK-PASS --output reports//CMP-WRK-DATA-QUEUE-DOWNLINK-PASS.json	{"finite_numeric_required":true,"scenario":"downlink_pass"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-DATA-QUEUE-OVERFLOW-PROTECTION	component	data_queue	normal_working / overflow_protection	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-DATA-QUEUE-OVERFLOW-PROTECTION --output reports//CMP-WRK-DATA-QUEUE-OVERFLOW-PROTECTION.json	{"finite_numeric_required":true,"scenario":"overflow_protection"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-ALL-DATA-QUEUE-RUN-ALL-MODES	component	data_queue	batch / run_all_modes	python scripts/run_engineering_verification_matrix.py --test-id CMP-ALL-DATA-QUEUE-RUN-ALL-MODES --output reports//CMP-ALL-DATA-QUEUE-RUN-ALL-MODES.json	{"minimum_top_level_modes":3,"finite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2-light coverage and finite numeric outputs.
CMP-FLT-DATA-QUEUE-OVERFLOW	component	data_queue	fault / overflow	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-DATA-QUEUE-OVERFLOW --output reports//CMP-FLT-DATA-QUEUE-OVERFLOW.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
CMP-FLT-DATA-QUEUE-READ-STALL	component	data_queue	fault / read_stall	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-DATA-QUEUE-READ-STALL --output reports//CMP-FLT-DATA-QUEUE-READ-STALL.json	{"active_fault_required":true,"sample_window_s":{0,600}}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-DEG-DATA-QUEUE-THROUGHPUT-DECAY	component	data_queue	degradation / throughput_decay	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-DATA-QUEUE-THROUGHPUT-DECAY --output reports//CMP-DEG-DATA-QUEUE-THROUGHPUT-DECAY.json	{"active_degradation_required":true,"sample_window_s":{0,600}}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-DEG-DATA-QUEUE-PACKET-ERROR-GROWTH	component	data_queue	degradation / packet_error_growth	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-DATA-QUEUE-PACKET-ERROR-GROWTH --output reports//CMP-DEG-DATA-QUEUE-PACKET-ERROR-GROWTH.json	{"active_degradation_required":true,"sample_window_s":{0,600}}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-L2-DATA-QUEUE-FAULT-DEFAULT-SET	component	data_queue	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-DATA-QUEUE-FAULT-DEFAULT-SET --output reports//CMP-L2-DATA-QUEUE-FAULT-DEFAULT-	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				SET.json		
CMP-L2-DATA-QUEUE-DEGRADATION-DEFAULT-SET	component	data_queue	degradation_default_set / degradation_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-DATA-QUEUE-DEGRADATION-DEFAULT-SET --output reports//CMP-L2-DATA-QUEUE-DEGRADATION-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default degradations participate at least once.
CMP-L2-DATA-QUEUE-COMBINED-DEFAULT-SET	component	data_queue	combined_default_set / combined_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-DATA-QUEUE-COMBINED-DEFAULT-SET --output reports//CMP-L2-DATA-QUEUE-COMBINED-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	At least one fault and one degradation participate in one run.
CMP-BND-DATA-QUEUE-SHORT-DURATION-60S	component	data_queue	duration_boundary / short_duration_60s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-DATA-QUEUE-SHORT-DURATION-60S --output reports//CMP-BND-DATA-QUEUE-SHORT-DURATION-60S.json	{"duration_s":60.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-BND-DATA-QUEUE-LONG-DURATION-86400S	component	data_queue	duration_boundary / long_duration_86400s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-DATA-QUEUE-LONG-DURATION-86400S --	{"duration_s":86400.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				output reports//CMP-BND-DATA-QUEUE-LONG-DURATION-86400S.json		
CMP-NEG-DATA-QUEUE-UNSUPPORTED-WORKING-SCENARIO	component	data_queue	negative / unsupported_working_scenario	python scripts/run_engineering_verification_matrix.py --test-id CMP-NEG-DATA-QUEUE-UNSUPPORTED-WORKING-SCENARIO --output reports//CMP-NEG-DATA-QUEUE-UNSUPPORTED-WORKING-SCENARIO.json	{"expected_exception":"ValueError"}	Unsupported working scenario is rejected by the component runner.
CMP-DET-DATA-QUEUE-REPEAT-STABILITY	component	data_queue	determinism / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id CMP-DET-DATA-QUEUE-REPEAT-STABILITY --output reports//CMP-DET-DATA-QUEUE-REPEAT-STABILITY.json	{"repeat_count":2,"semantic_hash_equal":true}	Two same-input runs produce the same semantic hash.
CMP-NOM-FUEL-TANK-NOMINAL	component	fuel_tank	nominal / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-NOM-FUEL-TANK-NOMINAL --output reports//CMP-NOM-FUEL-TANK-NOMINAL.json	{"finite_numeric_required":true,"error_key_forbidden":true}	run_nominal_case returns structured data without error and all numeric outputs are finite.
CMP-WRK-FUEL-TANK-NOMINAL	component	fuel_tank	normal_working / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-	{"finite_numeric_required":true,"scenario":"nominal"}	Named normal operating scenario executes and returns finite outputs with no error field.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				WRK-FUEL-TANK-NOMINAL --output reports//CMP-WRK-FUEL-TANK-NOMINAL.json		
CMP-WRK-FUEL-TANK-STANDBY-NO-FLOW	component	fuel_tank	normal_working / standby_no_flow	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-FUEL-TANK-STANDBY-NO-FLOW --output reports//CMP-WRK-FUEL-TANK-STANDBY-NO-FLOW.json	{"finite_numeric_required":true,"scenario":"standby_no_flow"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-FUEL-TANK-STATION-KEEPING-BURN	component	fuel_tank	normal_working / station_keeping_burn	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-FUEL-TANK-STATION-KEEPING-BURN --output reports//CMP-WRK-FUEL-TANK-STATION-KEEPING-BURN.json	{"finite_numeric_required":true,"scenario":"station_keeping_burn"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-FUEL-TANK-HIGH-RATE-BURN	component	fuel_tank	normal_working / high_rate_burn	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-FUEL-TANK-HIGH-RATE-BURN --output reports//CMP-WRK-FUEL-TANK-HIGH-RATE-BURN.json	{"finite_numeric_required":true,"scenario":"high_rate_burn"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-ALL-FUEL-TANK-RUN-ALL-MODES	component	fuel_tank	batch / run_all_modes	python scripts/run_engineering_verification_matrix.py --test-id CMP-	{"minimum_top_level_modes":3,"finite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2-light coverage and finite numeric outputs.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				ALL-FUEL-TANK-RUN-ALL-MODES --output reports//CMP-ALL-FUEL-TANK-RUN-ALL-MODES.json		
CMP-FLT-FUEL-TANK-LEAK	component	fuel_tank	fault / leak	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-FUEL-TANK-LEAK --output reports//CMP-FLT-FUEL-TANK-LEAK.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-FLT-FUEL-TANK-OUTLET-BLOCKAGE	component	fuel_tank	fault / outlet_blockage	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-FUEL-TANK-OUTLET-BLOCKAGE --output reports//CMP-FLT-FUEL-TANK-OUTLET-BLOCKAGE.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-DEG-FUEL-TANK-PRESSURE-DECAY	component	fuel_tank	degradation / pressure_decay	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-FUEL-TANK-PRESSURE-DECAY --output reports//CMP-DEG-FUEL-TANK-PRESSURE-DECAY.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-DEG-FUEL-TANK-OUTLET-AGING	component	fuel_tank	degradation / outlet_aging	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-FUEL-TANK-OUTLET-AGING --	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				output reports//CMP-DEG-FUEL-TANK-OUTLET-AGING.json		
CMP-L2-FUEL-TANK-FAULT-DEFAULT-SET	component	fuel_tank	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-FUEL-TANK-FAULT-DEFAULT-SET --output reports//CMP-L2-FUEL-TANK-FAULT-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.
CMP-L2-FUEL-TANK-DEGRADATION-DEFAULT-SET	component	fuel_tank	degradation_default_set / degradation_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-FUEL-TANK-DEGRADATION-DEFAULT-SET --output reports//CMP-L2-FUEL-TANK-DEGRADATION-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default degradations participate at least once.
CMP-L2-FUEL-TANK-COMBINED-DEFAULT-SET	component	fuel_tank	combined_default_set / combined_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-FUEL-TANK-COMBINED-DEFAULT-SET --output reports//CMP-L2-FUEL-TANK-COMBINED-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	At least one fault and one degradation participate in one run.
CMP-BND-FUEL-TANK-SHORT-DURATION-60S	component	fuel_tank	duration_boundary / short_duration_60s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-FUEL-TANK-SHORT-	{"duration_s":60.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				DURATION-60S -- output reports//CMP- BND-FUEL-TANK- SHORT- DURATION-60S.json		
CMP-BND-FUEL-TANK-LONG- DURATION-86400S	component	fuel_tank	duration_boundary / long_duration_86400s	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- BND-FUEL-TANK- LONG- DURATION-86400S -- output reports//CMP- BND-FUEL-TANK- LONG- DURATION-86400S.js on	{"duration_s":86400.0,"finite_nume ric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-NEG-FUEL-TANK- UNSUPPORTED-WORKING- SCENARIO	component	fuel_tank	negative / unsupported_working_scenario	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- NEG-FUEL-TANK- UNSUPPORTED- WORKING- SCENARIO --output reports//CMP-NEG- FUEL-TANK- UNSUPPORTED- WORKING- SCENARIO.json	{"expected_exception":"ValueError "}	Unsupported working scenario is rejected by the component runner.
CMP-DET-FUEL-TANK-REPEAT- STABILITY	component	fuel_tank	determinism / repeat_stability	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- DET-FUEL-TANK- REPEAT-STABILITY -- output reports//CMP- DET-FUEL-TANK- REPEAT- STABILITY.json	{"repeat_count":2,"semantic_hash_ equal":true}	Two same-input runs produce the same semantic hash.
CMP-NOM-GROUND-STATION-	component	ground_station	nominal / nominal	python	{"finite_numeric_required":true,"e	run_nominal_case returns

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
NOMINAL				scripts/run_engineering_verification_matrix.py --test-id CMP-NOM-GROUND-STATION-NOMINAL --output reports//CMP-NOM-GROUND-STATION-NOMINAL.json	error_key_forbidden:true}	structured data without error and all numeric outputs are finite.
CMP-WRK-GROUND-STATION-NOMINAL	component	ground_station	normal_working / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-GROUND-STATION-NOMINAL --output reports//CMP-WRK-GROUND-STATION-NOMINAL.json	{"finite_numeric_required":true,"scenario":"nominal"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-GROUND-STATION-OVERHEAD-PASS	component	ground_station	normal_working / overhead_pass	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-GROUND-STATION-OVERHEAD-PASS --output reports//CMP-WRK-GROUND-STATION-OVERHEAD-PASS.json	{"finite_numeric_required":true,"scenario":"overhead_pass"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-GROUND-STATION-HORIZON-MISS	component	ground_station	normal_working / horizon_miss	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-GROUND-STATION-HORIZON-MISS --output reports//CMP-WRK-GROUND-STATION-HORIZON-MISS.json	{"finite_numeric_required":true,"scenario":"horizon_miss"}	Named normal operating scenario executes and returns finite outputs with no error field.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
CMP-WRK-GROUND-STATION- STRICT-ELEVATION-MASK	component	ground_station	normal_working / strict_elevation_mask	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- WRK-GROUND- STATION-STRICT- ELEVATION-MASK -- output reports//CMP- WRK-GROUND- STATION-STRICT- ELEVATION- MASK.json	{"finite_numeric_required":true,"sc enario":"strict_elevation_mask"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-ALL-GROUND-STATION-RUN- ALL-MODES	component	ground_station	batch / run_all_modes	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- ALL-GROUND- STATION-RUN-ALL- MODES --output reports//CMP-ALL- GROUND-STATION- RUN-ALL- MODES.json	{"minimum_top_level_modes":3,"fi nite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2- light coverage and finite numeric outputs.
CMP-FLT-GROUND-STATION- TRACKING-LOSS	component	ground_station	fault / tracking_loss	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- FLT-GROUND- STATION-TRACKING- LOSS --output reports//CMP-FLT- GROUND-STATION- TRACKING-LOSS.json	{"active_fault_required":true,"sam ple_window_s":{0,600}}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-FLT-GROUND-STATION- WEATHER-FADE	component	ground_station	fault / weather_fade	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- FLT-GROUND- STATION-WEATHER- FADE --output reports//CMP-FLT-	{"active_fault_required":true,"sam ple_window_s":{0,600}}	The single fault is active in its configured time window and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				GROUND-STATION-WEATHER-FADE.json		
CMP-DEG-GROUND-STATION-GOVER-TDRIFT	component	ground_station	degradation / gover_tdrift	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-GROUND-STATION-GOVER-TDRIFT --output reports//CMP-DEG-GROUND-STATION-GOVER-TDRIFT.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-DEG-GROUND-STATION-POINTING-ACCURACY-DRIFT	component	ground_station	degradation / pointing_accuracy_drift	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-GROUND-STATION-POINTING-ACCURACY-DRIFT --output reports//CMP-DEG-GROUND-STATION-POINTING-ACCURACY-DRIFT.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-L2-GROUND-STATION-FAULT-DEFAULT-SET	component	ground_station	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-GROUND-STATION-FAULT-DEFAULT-SET --output reports//CMP-L2-GROUND-STATION-FAULT-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.
CMP-L2-GROUND-STATION-DEGRADATION-DEFAULT-SET	component	ground_station	degradation_default_set / degradation_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-GROUND-STATION-DEGRADATION-	{"duration_s":600,"finite_numeric_required":true}	All default degradations participate at least once.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				DEFAULT-SET -- output reports//CMP- L2-GROUND- STATION- DEGRADATION- DEFAULT-SET.json		
CMP-L2-GROUND-STATION- COMBINED-DEFAULT-SET	component	ground_station	combined_default_set / combined_default_set	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP-L2- GROUND-STATION- COMBINED- DEFAULT-SET -- output reports//CMP- L2-GROUND- STATION- COMBINED- DEFAULT-SET.json	{"duration_s":600,"finite_numeric_ required":true}	At least one fault and one degradation participate in one run.
CMP-BND-GROUND-STATION- SHORT-DURATION-60S	component	ground_station	duration_boundary / short_duration_60s	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- BND-GROUND- STATION-SHORT- DURATION-60S -- output reports//CMP- BND-GROUND- STATION-SHORT- DURATION-60S.json	{"duration_s":60.0,"finite_numeric_ required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-BND-GROUND-STATION- LONG-DURATION-86400S	component	ground_station	duration_boundary / long_duration_86400s	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- BND-GROUND- STATION-LONG- DURATION-86400S -- output reports//CMP- BND-GROUND- STATION-LONG- DURATION-86400S.js on	{"duration_s":86400.0,"finite_num eric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
CMP-NEG-GROUND-STATION-UNSUPPORTED-WORKING-SCENARIO	component	ground_station	negative / unsupported_working_scenario	python scripts/run_engineering_verification_matrix.py --test-id CMP-NEG-GROUND-STATION-UNSUPPORTED-WORKING-SCENARIO --output reports//CMP-NEG-GROUND-STATION-UNSUPPORTED-WORKING-SCENARIO.json	{"expected_exception":"ValueError"}	Unsupported working scenario is rejected by the component runner.
CMP-DET-GROUND-STATION-REPEAT-STABILITY	component	ground_station	determinism / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id CMP-DET-GROUND-STATION-REPEAT-STABILITY --output reports//CMP-DET-GROUND-STATION-REPEAT-STABILITY.json	{"repeat_count":2,"semantic_hash_equal":true}	Two same-input runs produce the same semantic hash.
CMP-NOM-HEATER-NOMINAL	component	heater	nominal / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-NOM-HEATER-NOMINAL --output reports//CMP-NOM-HEATER-NOMINAL.json	{"finite_numeric_required":true,"error_key_forbidden":true}	run_nominal_case returns structured data without error and all numeric outputs are finite.
CMP-ALL-HEATER-RUN-ALL-MODES	component	heater	batch / run_all_modes	python scripts/run_engineering_verification_matrix.py --test-id CMP-ALL-HEATER-RUN-ALL-MODES --output reports//CMP-ALL-	{"minimum_top_level_modes":3,"finite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2-light coverage and finite numeric outputs.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				HEATER-RUN-ALL-MODES.json		
CMP-FLT-HEATER-STUCK-OFF	component	heater	fault / stuck_off	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-HEATER-STUCK-OFF --output reports//CMP-FLT-HEATER-STUCK-OFF.json	{"active_fault_required":true,"sample_window_s":{0,600}}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-FLT-HEATER-STUCK-ON	component	heater	fault / stuck_on	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-HEATER-STUCK-ON --output reports//CMP-FLT-HEATER-STUCK-ON.json	{"active_fault_required":true,"sample_window_s":{0,600}}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-DEG-HEATER-HEATING-EFFICIENCY-DECAY	component	heater	degradation / heating_efficiency_decay	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-HEATER-HEATING-EFFICIENCY-DECAY --output reports//CMP-DEG-HEATER-HEATING-EFFICIENCY-DECAY.json	{"active_degradation_required":true,"sample_window_s":{0,600}}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-DEG-HEATER-RESISTANCE-DRIFT	component	heater	degradation / resistance_drift	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-HEATER-RESISTANCE-DRIFT --output reports//CMP-DEG-HEATER-RESISTANCE-	{"active_degradation_required":true,"sample_window_s":{0,600}}	The single degradation is active after onset and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				DRIFT.json		
CMP-L2-HEATER-FAULT-DEFAULT-SET	component	heater	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-HEATER-FAULT-DEFAULT-SET --output reports//CMP-L2-HEATER-FAULT-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.
CMP-L2-HEATER-DEGRADATION-DEFAULT-SET	component	heater	degradation_default_set / degradation_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-HEATER-DEGRADATION-DEFAULT-SET --output reports//CMP-L2-HEATER-DEGRADATION-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default degradations participate at least once.
CMP-L2-HEATER-COMBINED-DEFAULT-SET	component	heater	combined_default_set / combined_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-HEATER-COMBINED-DEFAULT-SET --output reports//CMP-L2-HEATER-COMBINED-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	At least one fault and one degradation participate in one run.
CMP-BND-HEATER-SHORT-DURATION-60S	component	heater	duration_boundary / short_duration_60s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-HEATER-SHORT-DURATION-60S --output reports//CMP-BND-HEATER-SHORT-	{"duration_s":600.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				DURATION-60S.json		
CMP-BND-HEATER-LONG-DURATION-86400S	component	heater	duration_boundary / long_duration_86400s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-HEATER-LONG-DURATION-86400S --output reports//CMP-BND-HEATER-LONG-DURATION-86400S.json	{"duration_s":86400.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-NEG-HEATER-UNSUPPORTED-WORKING-SCENARIO	component	heater	negative / unsupported_working_scenario	python scripts/run_engineering_verification_matrix.py --test-id CMP-NEG-HEATER-UNSUPPORTED-WORKING-SCENARIO --output reports//CMP-NEG-HEATER-UNSUPPORTED-WORKING-SCENARIO.json	{"expected_exception":"ValueError"}	Unsupported working scenario is rejected by the component runner.
CMP-DET-HEATER-REPEAT-STABILITY	component	heater	determinism / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id CMP-DET-HEATER-REPEAT-STABILITY --output reports//CMP-DET-HEATER-REPEAT-STABILITY.json	{"repeat_count":2,"semantic_hash_equal":true}	Two same-input runs produce the same semantic hash.
CMP-NOM-IMU-NOMINAL	component	imu	nominal / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-NOM-IMU-NOMINAL --output reports//CMP-NOM-	{"finite_numeric_required":true,"error_key_forbidden":true}	run_nominal_case returns structured data without error and all numeric outputs are finite.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				IMU-NOMINAL.json		
CMP-WRK-IMU-NOMINAL	component	imu	normal_working / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-IMU-NOMINAL --output reports//CMP-WRK-IMU-NOMINAL.json	{"finite_numeric_required":true,"scenario":"nominal"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-IMU-SHORT-BIAS-WALK	component	imu	normal_working / short_bias_walk	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-IMU-SHORT-BIAS-WALK --output reports//CMP-WRK-IMU-SHORT-BIAS-WALK.json	{"finite_numeric_required":true,"scenario":"short_bias_walk"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-IMU-LONG-BIAS-WALK	component	imu	normal_working / long_bias_walk	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-IMU-LONG-BIAS-WALK --output reports//CMP-WRK-IMU-LONG-BIAS-WALK.json	{"finite_numeric_required":true,"scenario":"long_bias_walk"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-IMU-MEASUREMENT-READOUT	component	imu	normal_working / measurement_readout	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-IMU-MEASUREMENT-READOUT --output reports//CMP-WRK-IMU-MEASUREMENT-READOUT.json	{"finite_numeric_required":true,"scenario":"measurement_readout"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-ALL-IMU-RUN-ALL-MODES	component	imu	batch / run_all_modes	python scripts/run_engineering_verification_matrix.py	{"minimum_top_level_modes":3,"finite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2-light coverage and finite numeric outputs.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				x.py --test-id CMP-ALL-IMU-RUN-ALL-MODES --output reports//CMP-ALL-IMU-RUN-ALL-MODES.json		
CMP-FLT-IMU-BIAS-STEP	component	imu	fault / bias_step	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-IMU-BIAS-STEP --output reports//CMP-FLT-IMU-BIAS-STEP.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-FLT-IMU-AXIS-DROPOUT	component	imu	fault / axis_dropout	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-IMU-AXIS-DROPOUT --output reports//CMP-FLT-IMU-AXIS-DROPOUT.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-DEG-IMU-BIAS-DRIFT	component	imu	degradation / bias_drift	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-IMU-BIAS-DRIFT --output reports//CMP-DEG-IMU-BIAS-DRIFT.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-DEG-IMU-NOISE-DENSITY-GROWTH	component	imu	degradation / noise_density_growth	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-IMU-NOISE-DENSITY-GROWTH --output reports//CMP-DEG-IMU-NOISE-DENSITY-GROWTH.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
CMP-L2-IMU-FAULT-DEFAULT-SET	component	imu	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-IMU-FAULT-DEFAULT-SET --output reports//CMP-L2-IMU-FAULT-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.
CMP-L2-IMU-DEGRADATION-DEFAULT-SET	component	imu	degradation_default_set / degradation_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-IMU-DEGRADATION-DEFAULT-SET --output reports//CMP-L2-IMU-DEGRADATION-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default degradations participate at least once.
CMP-L2-IMU-COMBINED-DEFAULT-SET	component	imu	combined_default_set / combined_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-IMU-COMBINED-DEFAULT-SET --output reports//CMP-L2-IMU-COMBINED-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	At least one fault and one degradation participate in one run.
CMP-BND-IMU-SHORT-DURATION-60S	component	imu	duration_boundary / short_duration_60s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-IMU-SHORT-DURATION-60S --output reports//CMP-BND-IMU-SHORT-DURATION-60S.json	{"duration_s":60.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-BND-IMU-LONG-DURATION-86400S	component	imu	duration_boundary / long_duration_86400s	python scripts/run_engineering_verification_matrix.py --test-id CMP-	{"duration_s":86400.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				BND-IMU-LONG-DURATION-86400S --output reports//CMP-BND-IMU-LONG-DURATION-86400S.json		
CMP-NEG-IMU-UNSUPPORTED-WORKING-SCENARIO	component	imu	negative / unsupported_working_scenario	python scripts/run_engineering_verification_matrix.py --test-id CMP-NEG-IMU-UNSUPPORTED-WORKING-SCENARIO --output reports//CMP-NEG-IMU-UNSUPPORTED-WORKING-SCENARIO.json	{"expected_exception":"ValueError"}	Unsupported working scenario is rejected by the component runner.
CMP-DET-IMU-REPEAT-STABILITY	component	imu	determinism / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id CMP-DET-IMU-REPEAT-STABILITY --output reports//CMP-DET-IMU-REPEAT-STABILITY.json	{"repeat_count":2,"semantic_hash_equal":true}	Two same-input runs produce the same semantic hash.
CMP-NOM-LINK-BUDGET-NOMINAL	component	link_budget	nominal / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-NOM-LINK-BUDGET-NOMINAL --output reports//CMP-NOM-LINK-BUDGET-NOMINAL.json	{"finite_numeric_required":true,"error_key_forbidden":true}	run_nominal_case returns structured data without error and all numeric outputs are finite.
CMP-WRK-LINK-BUDGET-NOMINAL	component	link_budget	normal_working / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-LINK-BUDGET-	{"finite_numeric_required":true,"scenario":"nominal"}	Named normal operating scenario executes and returns finite outputs with no error field.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				NOMINAL --output reports//CMP-WRK-LINK-BUDGET-NOMINAL.json		
CMP-WRK-LINK-BUDGET-NEAR-CONTACT	component	link_budget	normal_working / near_contact	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-LINK-BUDGET-NEAR-CONTACT --output reports//CMP-WRK-LINK-BUDGET-NEAR-CONTACT.json	{"finite_numeric_required":true,"scenario":"near_contact"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-LINK-BUDGET-ORBIT-RANGE-SWEEP	component	link_budget	normal_working / orbit_range_sweep	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-LINK-BUDGET-ORBIT-RANGE-SWEEP --output reports//CMP-WRK-LINK-BUDGET-ORBIT-RANGE-SWEEP.json	{"finite_numeric_required":true,"scenario":"orbit_range_sweep"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-LINK-BUDGET-LOW-POWER-DOWNLINK	component	link_budget	normal_working / low_power_downlink	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-LINK-BUDGET-LOW-POWER-DOWNLINK --output reports//CMP-WRK-LINK-BUDGET-LOW-POWER-DOWNLINK.json	{"finite_numeric_required":true,"scenario":"low_power_downlink"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-ALL-LINK-BUDGET-RUN-ALL-MODES	component	link_budget	batch / run_all_modes	python scripts/run_engineering_verification_matrix.py --test-id CMP-ALL-LINK-BUDGET-RUN-ALL-MODES --	{"minimum_top_level_modes":3,"finite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2-light coverage and finite numeric outputs.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				output reports//CMP-ALL-LINK-BUDGET-RUN-ALL-MODES.json		
CMP-FLT-LINK-BUDGET-JAMMING	component	link_budget	fault / jamming	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-LINK-BUDGET-JAMMING --output reports//CMP-FLT-LINK-BUDGET-JAMMING.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-FLT-LINK-BUDGET-FADE	component	link_budget	fault / fade	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-LINK-BUDGET-FADE --output reports//CMP-FLT-LINK-BUDGET-FADE.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-DEG-LINK-BUDGET-MARGIN-EROSION	component	link_budget	degradation / margin_erosion	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-LINK-BUDGET-MARGIN-EROSION --output reports//CMP-DEG-LINK-BUDGET-MARGIN-EROSION.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-DEG-LINK-BUDGET-NOISE-FIGURE-GROWTH	component	link_budget	degradation / noise_figure_growth	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-LINK-BUDGET-NOISE-FIGURE-GROWTH --output reports//CMP-DEG-LINK-BUDGET-	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				NOISE-FIGURE-GROWTH.json		
CMP-L2-LINK-BUDGET-FAULT-DEFAULT-SET	component	link_budget	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-LINK-BUDGET-FAULT-DEFAULT-SET --output reports//CMP-L2-LINK-BUDGET-FAULT-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.
CMP-L2-LINK-BUDGET-DEGRADATION-DEFAULT-SET	component	link_budget	degradation_default_set / degradation_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-LINK-BUDGET-DEGRADATION-DEFAULT-SET --output reports//CMP-L2-LINK-BUDGET-DEGRADATION-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default degradations participate at least once.
CMP-L2-LINK-BUDGET-COMBINED-DEFAULT-SET	component	link_budget	combined_default_set / combined_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-LINK-BUDGET-COMBINED-DEFAULT-SET --output reports//CMP-L2-LINK-BUDGET-COMBINED-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	At least one fault and one degradation participate in one run.
CMP-BND-LINK-BUDGET-SHORT-DURATION-60S	component	link_budget	duration_boundary / short_duration_60s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-LINK-BUDGET-SHORT-	{"duration_s":60.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				DURATION-60S -- output reports//CMP- BND-LINK-BUDGET- SHORT- DURATION-60S.json		
CMP-BND-LINK-BUDGET-LONG- DURATION-86400S	component	link_budget	duration_boundary / long_duration_86400s	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- BND-LINK-BUDGET- LONG- DURATION-86400S -- output reports//CMP- BND-LINK-BUDGET- LONG- DURATION-86400S.js on	{"duration_s":86400.0,"finite_nume ric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-NEG-LINK-BUDGET- UNSUPPORTED-WORKING- SCENARIO	component	link_budget	negative / unsupported_working_scenario	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- NEG-LINK-BUDGET- UNSUPPORTED- WORKING- SCENARIO --output reports//CMP-NEG- LINK-BUDGET- UNSUPPORTED- WORKING- SCENARIO.json	{"expected_exception":"ValueError "}	Unsupported working scenario is rejected by the component runner.
CMP-DET-LINK-BUDGET-REPEAT- STABILITY	component	link_budget	determinism / repeat_stability	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- DET-LINK-BUDGET- REPEAT-STABILITY -- output reports//CMP- DET-LINK-BUDGET- REPEAT- STABILITY.json	{"repeat_count":2,"semantic_hash_ equal":true}	Two same-input runs produce the same semantic hash.
CMP-NOM-MAGNETOMETER-	component	magnetometer	nominal / nominal	python	{"finite_numeric_required":true,"e	run_nominal_case returns

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
NOMINAL				scripts/run_engineering_verification_matrix.py --test-id CMP-NOM-MAGNETOMETER-NOMINAL --output reports//CMP-NOM-MAGNETOMETER-NOMINAL.json	error_key_forbidden:true}	structured data without error and all numeric outputs are finite.
CMP-WRK-MAGNETOMETER-NOMINAL	component	magnetometer	normal_working / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-MAGNETOMETER-NOMINAL --output reports//CMP-WRK-MAGNETOMETER-NOMINAL.json	{"finite_numeric_required":true,"scenario":"nominal"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-MAGNETOMETER-BODY-FRAME-MEASUREMENT	component	magnetometer	normal_working / body_frame_measurement	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-MAGNETOMETER-BODY-FRAME-MEASUREMENT --output reports//CMP-WRK-MAGNETOMETER-BODY-FRAME-MEASUREMENT.json	{"finite_numeric_required":true,"scenario":"body_frame_measurement"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-MAGNETOMETER-ROTATED-MOUNTING	component	magnetometer	normal_working / rotated_mounting	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-MAGNETOMETER-ROTATED-MOUNTING --output reports//CMP-WRK-	{"finite_numeric_required":true,"scenario":"rotated_mounting"}	Named normal operating scenario executes and returns finite outputs with no error field.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				MAGNETOMETER-ROTATED-MOUNTING.json		
CMP-WRK-MAGNETOMETER-BIASED-READOUT	component	magnetometer	normal_working / biased_readout	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-MAGNETOMETER-BIASED-READOUT --output reports//CMP-WRK-MAGNETOMETER-BIASED-READOUT.json	{"finite_numeric_required":true,"scenario":"biased_readout"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-ALL-MAGNETOMETER-RUN-ALL-MODES	component	magnetometer	batch / run_all_modes	python scripts/run_engineering_verification_matrix.py --test-id CMP-ALL-MAGNETOMETER-RUN-ALL-MODES --output reports//CMP-ALL-MAGNETOMETER-RUN-ALL-MODES.json	{"minimum_top_level_modes":3,"finite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2-light coverage and finite numeric outputs.
CMP-FLT-MAGNETOMETER-AXIS-BIAS	component	magnetometer	fault / axis_bias	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-MAGNETOMETER-AXIS-BIAS --output reports//CMP-FLT-MAGNETOMETER-AXIS-BIAS.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-FLT-MAGNETOMETER-AXIS-DROPOUT	component	magnetometer	fault / axis_dropout	python scripts/run_engineering_verification_matrix.py --test-id CMP-	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				FLT-MAGNETOMETER-AXIS-DROPOUT --output reports//CMP-FLT-MAGNETOMETER-AXIS-DROPOUT.json		
CMP-DEG-MAGNETOMETER-BIAS-DRIFT	component	magnetometer	degradation / bias_drift	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-MAGNETOMETER-BIAS-DRIFT --output reports//CMP-DEG-MAGNETOMETER-BIAS-DRIFT.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-DEG-MAGNETOMETER-SCALE-FACTOR-DRIFT	component	magnetometer	degradation / scale_factor_drift	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-MAGNETOMETER-SCALE-FACTOR-DRIFT --output reports//CMP-DEG-MAGNETOMETER-SCALE-FACTOR-DRIFT.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-L2-MAGNETOMETER-FAULT-DEFAULT-SET	component	magnetometer	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-MAGNETOMETER-FAULT-DEFAULT-SET --output reports//CMP-L2-MAGNETOMETER-FAULT-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.
CMP-L2-MAGNETOMETER-	component	magnetometer	degradation_default_set /	python	{"duration_s":600,"finite_numeric_	All default degradations

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
DEGRADATION-DEFAULT-SET			degradation_default_set	scripts/run_engineering_verification_matrix.py --test-id CMP-L2-MAGNETOMETER-DEGRADATION-DEFAULT-SET --output reports//CMP-L2-MAGNETOMETER-DEGRADATION-DEFAULT-SET.json	required":true}	participate at least once.
CMP-L2-MAGNETOMETER-COMBINED-DEFAULT-SET	component	magnetometer	combined_default_set / combined_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-MAGNETOMETER-COMBINED-DEFAULT-SET --output reports//CMP-L2-MAGNETOMETER-COMBINED-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	At least one fault and one degradation participate in one run.
CMP-BND-MAGNETOMETER-SHORT-DURATION-60S	component	magnetometer	duration_boundary / short_duration_60s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-MAGNETOMETER-SHORT-DURATION-60S --output reports//CMP-BND-MAGNETOMETER-SHORT-DURATION-60S.json	{"duration_s":60.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-BND-MAGNETOMETER-LONG-DURATION-86400S	component	magnetometer	duration_boundary / long_duration_86400s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-	{"duration_s":86400.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				MAGNETOMETER-LONG-DURATION-86400S --output reports//CMP-BND-MAGNETOMETER-LONG-DURATION-86400S.js on		
CMP-NEG-MAGNETOMETER-UNSUPPORTED-WORKING-SCENARIO	component	magnetometer	negative / unsupported_working_scenario	python scripts/run_engineering_verification_matrix.py --test-id CMP-NEG-MAGNETOMETER-UNSUPPORTED-WORKING-SCENARIO --output reports//CMP-NEG-MAGNETOMETER-UNSUPPORTED-WORKING-SCENARIO.json	{"expected_exception":"ValueError"}	Unsupported working scenario is rejected by the component runner.
CMP-DET-MAGNETOMETER-REPEAT-STABILITY	component	magnetometer	determinism / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id CMP-DET-MAGNETOMETER-REPEAT-STABILITY --output reports//CMP-DET-MAGNETOMETER-REPEAT-STABILITY.json	{"repeat_count":2,"semantic_hash_equal":true}	Two same-input runs produce the same semantic hash.
CMP-NOM-MTB-NOMINAL	component	mtb	nominal / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-NOM-MTB-NOMINAL --output	{"finite_numeric_required":true,"error_key_forbidden":true}	run_nominal_case returns structured data without error and all numeric outputs are finite.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				reports//CMP-NOM-MTB-NOMINAL.json		
CMP-WRK-MTB-NOMINAL	component	mtb	normal_working / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-MTB-NOMINAL --output reports//CMP-WRK-MTB-NOMINAL.json	{"finite_numeric_required":true,"scenario":"nominal"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-MTB-SINGLE-AXIS-DIPOLE	component	mtb	normal_working / single_axis_dipole	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-MTB-SINGLE-AXIS-DIPOLE --output reports//CMP-WRK-MTB-SINGLE-AXIS-DIPOLE.json	{"finite_numeric_required":true,"scenario":"single_axis_dipole"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-MTB-AXIS-REVERSAL	component	mtb	normal_working / axis_reversal	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-MTB-AXIS-REVERSAL --output reports//CMP-WRK-MTB-AXIS-REVERSAL.json	{"finite_numeric_required":true,"scenario":"axis_reversal"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-MTB-SATURATED-COMMAND	component	mtb	normal_working / saturated_command	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-MTB-SATURATED-COMMAND --output reports//CMP-WRK-MTB-SATURATED-COMMAND.json	{"finite_numeric_required":true,"scenario":"saturated_command"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-ALL-MTB-RUN-ALL-MODES	component	mtb	batch / run_all_modes	python scripts/run_engineering_verification_matrix.py	{"minimum_top_level_modes":3,"finite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2-light coverage and finite numeric outputs.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				x.py --test-id CMP-ALL-MTB-RUN-ALL-MODES --output reports//CMP-ALL-MTB-RUN-ALL-MODES.json		
CMP-FLT-MTB-COIL-OPEN	component	mtb	fault / coil_open	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-MTB-COIL-OPEN --output reports//CMP-FLT-MTB-COIL-OPEN.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-FLT-MTB-COIL-SHORT	component	mtb	fault / coil_short	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-MTB-COIL-SHORT --output reports//CMP-FLT-MTB-COIL-SHORT.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-DEG-MTB-COIL-RESISTANCE-GROWTH	component	mtb	degradation / coil_resistance_growth	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-MTB-COIL-RESISTANCE-GROWTH --output reports//CMP-DEG-MTB-COIL-RESISTANCE-GROWTH.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-DEG-MTB-MAGNETIC-MOMENT-DRIFT	component	mtb	degradation / magnetic_moment_drift	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-MTB-MAGNETIC-MOMENT-DRIFT --	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				output reports//CMP-DEG-MTB-MAGNETIC-MOMENT-DRIFT.json		
CMP-L2-MTB-FAULT-DEFAULT-SET	component	mtb	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-MTB-FAULT-DEFAULT-SET --output reports//CMP-L2-MTB-FAULT-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.
CMP-L2-MTB-DEGRADATION-DEFAULT-SET	component	mtb	degradation_default_set / degradation_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-MTB-DEGRADATION-DEFAULT-SET --output reports//CMP-L2-MTB-DEGRADATION-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default degradations participate at least once.
CMP-L2-MTB-COMBINED-DEFAULT-SET	component	mtb	combined_default_set / combined_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-MTB-COMBINED-DEFAULT-SET --output reports//CMP-L2-MTB-COMBINED-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	At least one fault and one degradation participate in one run.
CMP-BND-MTB-SHORT-DURATION-60S	component	mtb	duration_boundary / short_duration_60s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-MTB-SHORT-DURATION-60S --output reports//CMP-BND-MTB-SHORT-DURATION-60S.json	{"duration_s":60.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
CMP-BND-MTB-LONG-DURATION-86400S	component	mtb	duration_boundary / long_duration_86400s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-MTB-LONG-DURATION-86400S --output reports//CMP-BND-MTB-LONG-DURATION-86400S.json	{"duration_s":86400.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-NEG-MTB-UNSUPPORTED-WORKING-SCENARIO	component	mtb	negative / unsupported_working_scenario	python scripts/run_engineering_verification_matrix.py --test-id CMP-NEG-MTB-UNSUPPORTED-WORKING-SCENARIO --output reports//CMP-NEG-MTB-UNSUPPORTED-WORKING-SCENARIO.json	{"expected_exception":"ValueError"}	Unsupported working scenario is rejected by the component runner.
CMP-DET-MTB-REPEAT-STABILITY	component	mtb	determinism / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id CMP-DET-MTB-REPEAT-STABILITY --output reports//CMP-DET-MTB-REPEAT-STABILITY.json	{"repeat_count":2,"semantic_hash_equal":true}	Two same-input runs produce the same semantic hash.
CMP-NOM-ONBOARD-STORAGE-NOMINAL	component	onboard_storage	nominal / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-NOM-ONBOARD-STORAGE-NOMINAL --output reports//CMP-NOM-ONBOARD-STORAGE-NOMINAL.json	{"finite_numeric_required":true,"error_key_forbidden":true}	run_nominal_case returns structured data without error and all numeric outputs are finite.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
CMP-WRK-ONBOARD-STORAGE-NOMINAL	component	onboard_storage	normal_working / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-ONBOARD-STORAGE-NOMINAL --output reports//CMP-WRK-ONBOARD-STORAGE-NOMINAL.json	{"finite_numeric_required":true,"scenario":"nominal"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-ONBOARD-STORAGE-PAYLOAD-RECORDING	component	onboard_storage	normal_working / payload_recording	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-ONBOARD-STORAGE-PAYLOAD-RECORDING --output reports//CMP-WRK-ONBOARD-STORAGE-PAYLOAD-RECORDING.json	{"finite_numeric_required":true,"scenario":"payload_recording"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-ONBOARD-STORAGE-DOWNLINK-DRAIN	component	onboard_storage	normal_working / downlink_drain	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-ONBOARD-STORAGE-DOWNLINK-DRAIN --output reports//CMP-WRK-ONBOARD-STORAGE-DOWNLINK-DRAIN.json	{"finite_numeric_required":true,"scenario":"downlink_drain"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-ONBOARD-STORAGE-STORAGE-OVERFLOW	component	onboard_storage	normal_working / storage_overflow	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-ONBOARD-STORAGE-STORAGE-OVERFLOW --output reports//CMP-WRK-	{"finite_numeric_required":true,"scenario":"storage_overflow"}	Named normal operating scenario executes and returns finite outputs with no error field.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				ONBOARD-STORAGE-STORAGE-OVERFLOW.json		
CMP-ALL-ONBOARD-STORAGE-RUN-ALL-MODES	component	onboard_storage	batch / run_all_modes	python scripts/run_engineering_verification_matrix.py --test-id CMP-ALL-ONBOARD-STORAGE-RUN-ALL-MODES --output reports//CMP-ALL-ONBOARD-STORAGE-RUN-ALL-MODES.json	{"minimum_top_level_modes":3,"finite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2-light coverage and finite numeric outputs.
CMP-FLT-ONBOARD-STORAGE-BAD-BLOCK	component	onboard_storage	fault / bad_block	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-ONBOARD-STORAGE-BAD-BLOCK --output reports//CMP-FLT-ONBOARD-STORAGE-BAD-BLOCK.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-FLT-ONBOARD-STORAGE-READ-ONLY	component	onboard_storage	fault / read_only	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-ONBOARD-STORAGE-READ-ONLY --output reports//CMP-FLT-ONBOARD-STORAGE-READ-ONLY.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-DEG-ONBOARD-STORAGE-CAPACITY-LOSS	component	onboard_storage	degradation / capacity_loss	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-ONBOARD-STORAGE-CAPACITY-LOSS --output	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				reports//CMP-DEG-ONBOARD-STORAGE-CAPACITY-LOSS.json		
CMP-DEG-ONBOARD-STORAGE-BIT-ERROR-GROWTH	component	onboard_storage	degradation / bit_error_growth	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-ONBOARD-STORAGE-BIT-ERROR-GROWTH --output reports//CMP-DEG-ONBOARD-STORAGE-BIT-ERROR-GROWTH.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-L2-ONBOARD-STORAGE-FAULT-DEFAULT-SET	component	onboard_storage	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-ONBOARD-STORAGE-FAULT-DEFAULT-SET --output reports//CMP-L2-ONBOARD-STORAGE-FAULT-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.
CMP-L2-ONBOARD-STORAGE-DEGRADATION-DEFAULT-SET	component	onboard_storage	degradation_default_set / degradation_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-ONBOARD-STORAGE-DEGRADATION-DEFAULT-SET --output reports//CMP-L2-ONBOARD-STORAGE-DEGRADATION-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default degradations participate at least once.
CMP-L2-ONBOARD-STORAGE-COMBINED-DEFAULT-SET	component	onboard_storage	combined_default_set / combined_default_set	python scripts/run_engineering_verification_matrix.py	{"duration_s":600,"finite_numeric_required":true}	At least one fault and one degradation participate in one run.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				x.py --test-id CMP-L2-ONBOARD-STORAGE-COMBINED-DEFAULT-SET --output reports//CMP-L2-ONBOARD-STORAGE-COMBINED-DEFAULT-SET.json		
CMP-BND-ONBOARD-STORAGE-SHORT-DURATION-60S	component	onboard_storage	duration_boundary / short_duration_60s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-ONBOARD-STORAGE-SHORT-DURATION-60S --output reports//CMP-BND-ONBOARD-STORAGE-SHORT-DURATION-60S.json	{"duration_s":60.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-BND-ONBOARD-STORAGE-LONG-DURATION-86400S	component	onboard_storage	duration_boundary / long_duration_86400s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-ONBOARD-STORAGE-LONG-DURATION-86400S --output reports//CMP-BND-ONBOARD-STORAGE-LONG-DURATION-86400S.json	{"duration_s":86400.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-NEG-ONBOARD-STORAGE-UNSUPPORTED-WORKING-SCENARIO	component	onboard_storage	negative / unsupported_working_scenario	python scripts/run_engineering_verification_matrix.py --test-id CMP-NEG-ONBOARD-STORAGE-UNSUPPORTED-WORKING-SCENARIO --output	{"expected_exception":"ValueError"}	Unsupported working scenario is rejected by the component runner.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				reports//CMP-NEG-ONBOARD-STORAGE-UNSUPPORTED-WORKING-SCENARIO.json		
CMP-DET-ONBOARD-STORAGE-REPEAT-STABILITY	component	onboard_storage	determinism / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id CMP-DET-ONBOARD-STORAGE-REPEAT-STABILITY --output reports//CMP-DET-ONBOARD-STORAGE-REPEAT-STABILITY.json	{"repeat_count":2,"semantic_hash_equal":true}	Two same-input runs produce the same semantic hash.
CMP-NOM-PAYLOAD-NOMINAL	component	payload	nominal / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-NOM-PAYLOAD-NOMINAL --output reports//CMP-NOM-PAYLOAD-NOMINAL.json	{"finite_numeric_required":true,"error_key_forbidden":true}	run_nominal_case returns structured data without error and all numeric outputs are finite.
CMP-WRK-PAYLOAD-NOMINAL	component	payload	normal_working / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-PAYLOAD-NOMINAL --output reports//CMP-WRK-PAYLOAD-NOMINAL.json	{"finite_numeric_required":true,"scenario":"nominal"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-PAYLOAD-OBSERVATION-PASS	component	payload	normal_working / observation_pass	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-PAYLOAD-OBSERVATION-PASS --output	{"finite_numeric_required":true,"scenario":"observation_pass"}	Named normal operating scenario executes and returns finite outputs with no error field.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				reports//CMP-WRK-PAYLOAD-OBSERVATION-PASS.json		
CMP-WRK-PAYLOAD-STANDBY-WINDOW	component	payload	normal_working / standby_window	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-PAYLOAD-STANDBY-WINDOW --output reports//CMP-WRK-PAYLOAD-STANDBY-WINDOW.json	{"finite_numeric_required":true,"scenario":"standby_window"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-PAYLOAD-POINTING-LIMITED	component	payload	normal_working / pointing_limited	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-PAYLOAD-POINTING-LIMITED --output reports//CMP-WRK-PAYLOAD-POINTING-LIMITED.json	{"finite_numeric_required":true,"scenario":"pointing_limited"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-ALL-PAYLOAD-RUN-ALL-MODES	component	payload	batch / run_all_modes	python scripts/run_engineering_verification_matrix.py --test-id CMP-ALL-PAYLOAD-RUN-ALL-MODES --output reports//CMP-ALL-PAYLOAD-RUN-ALL-MODES.json	{"minimum_top_level_modes":3,"finite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2-light coverage and finite numeric outputs.
CMP-FLT-PAYLOAD-INSTRUMENT-OFF	component	payload	fault / instrument_off	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-PAYLOAD-INSTRUMENT-OFF --output reports//CMP-FLT-PAYLOAD-	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				INSTRUMENT-OFF.json		
CMP-FLT-PAYLOAD-SATURATION	component	payload	fault / saturation	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-PAYLOAD-SATURATION --output reports//CMP-FLT-PAYLOAD-SATURATION.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-DEG-PAYLOAD-SENSITIVITY-DECAY	component	payload	degradation / sensitivity_decay	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-PAYLOAD-SENSITIVITY-DECAY --output reports//CMP-DEG-PAYLOAD-SENSITIVITY-DECAY.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-DEG-PAYLOAD-DARK-CURRENT-GROWTH	component	payload	degradation / dark_current_growth	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-PAYLOAD-DARK-CURRENT-GROWTH --output reports//CMP-DEG-PAYLOAD-DARK-CURRENT-GROWTH.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-L2-PAYLOAD-FAULT-DEFAULT-SET	component	payload	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-PAYLOAD-FAULT-DEFAULT-SET --output reports//CMP-L2-PAYLOAD-FAULT-	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				DEFAULT-SET.json		
CMP-L2-PAYLOAD-DEGRADATION-DEFAULT-SET	component	payload	degradation_default_set / degradation_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-PAYLOAD-DEGRADATION-DEFAULT-SET --output reports//CMP-L2-PAYLOAD-DEGRADATION-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default degradations participate at least once.
CMP-L2-PAYLOAD-COMBINED-DEFAULT-SET	component	payload	combined_default_set / combined_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-PAYLOAD-COMBINED-DEFAULT-SET --output reports//CMP-L2-PAYLOAD-COMBINED-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	At least one fault and one degradation participate in one run.
CMP-BND-PAYLOAD-SHORT-DURATION-60S	component	payload	duration_boundary / short_duration_60s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-PAYLOAD-SHORT-DURATION-60S --output reports//CMP-BND-PAYLOAD-SHORT-DURATION-60S.json	{"duration_s":60.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-BND-PAYLOAD-LONG-DURATION-86400S	component	payload	duration_boundary / long_duration_86400s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-PAYLOAD-LONG-DURATION-86400S --	{"duration_s":86400.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				output reports//CMP-BND-PAYLOAD-LONG-DURATION-86400S.json		
CMP-NEG-PAYLOAD-UNSUPPORTED-WORKING-SCENARIO	component	payload	negative / unsupported_working_scenario	python scripts/run_engineering_verification_matrix.py --test-id CMP-NEG-PAYLOAD-UNSUPPORTED-WORKING-SCENARIO --output reports//CMP-NEG-PAYLOAD-UNSUPPORTED-WORKING-SCENARIO.json	{"expected_exception":"ValueError"}	Unsupported working scenario is rejected by the component runner.
CMP-DET-PAYLOAD-REPEAT-STABILITY	component	payload	determinism / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id CMP-DET-PAYLOAD-REPEAT-STABILITY --output reports//CMP-DET-PAYLOAD-REPEAT-STABILITY.json	{"repeat_count":2,"semantic_hash_equal":true}	Two same-input runs produce the same semantic hash.
CMP-NOM-PAYLOAD-SENSOR-NOMINAL	component	payload_sensor	nominal / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-NOM-PAYLOAD-SENSOR-NOMINAL --output reports//CMP-NOM-PAYLOAD-SENSOR-NOMINAL.json	{"finite_numeric_required":true,"error_key_forbidden":true}	run_nominal_case returns structured data without error and all numeric outputs are finite.
CMP-WRK-PAYLOAD-SENSOR-NOMINAL	component	payload_sensor	normal_working / nominal	python scripts/run_engineering_verification_matrix.py	{"finite_numeric_required":true,"scenario":"nominal"}	Named normal operating scenario executes and returns finite outputs with no error field.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				x.py --test-id CMP-WRK-PAYLOAD-SENSOR-NOMINAL --output reports//CMP-WRK-PAYLOAD-SENSOR-NOMINAL.json		
CMP-WRK-PAYLOAD-SENSOR-CLEAR-NADIR	component	payload_sensor	normal_working / clear_nadir	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-PAYLOAD-SENSOR-CLEAR-NADIR --output reports//CMP-WRK-PAYLOAD-SENSOR-CLEAR-NADIR.json	{"finite_numeric_required":true,"scenario":"clear_nadir"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-PAYLOAD-SENSOR-CLOUDY-SCENE	component	payload_sensor	normal_working / cloudy_scene	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-PAYLOAD-SENSOR-CLOUDY-SCENE --output reports//CMP-WRK-PAYLOAD-SENSOR-CLOUDY-SCENE.json	{"finite_numeric_required":true,"scenario":"cloudy_scene"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-PAYLOAD-SENSOR-POINTING-LIMIT	component	payload_sensor	normal_working / pointing_limit	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-PAYLOAD-SENSOR-POINTING-LIMIT --output reports//CMP-WRK-PAYLOAD-SENSOR-POINTING-LIMIT.json	{"finite_numeric_required":true,"scenario":"pointing_limit"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-ALL-PAYLOAD-SENSOR-RUN-ALL-MODES	component	payload_sensor	batch / run_all_modes	python scripts/run_engineering_verification_matrix.py	{"minimum_top_level_modes":3,"finite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2-light coverage and finite numeric outputs.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				x.py --test-id CMP-ALL-PAYLOAD-SENSOR-RUN-ALL-MODES --output reports//CMP-ALL-PAYLOAD-SENSOR-RUN-ALL-MODES.json		
CMP-FLT-PAYLOAD-SENSOR-PIXEL-DROPOUT	component	payload_sensor	fault / pixel_dropout	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-PAYLOAD-SENSOR-PIXEL-DROPOUT --output reports//CMP-FLT-PAYLOAD-SENSOR-PIXEL-DROPOUT.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-FLT-PAYLOAD-SENSOR-BLUR	component	payload_sensor	fault / blur	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-PAYLOAD-SENSOR-BLUR --output reports//CMP-FLT-PAYLOAD-SENSOR-BLUR.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-DEG-PAYLOAD-SENSOR-RESPONSIVITY-DECAY	component	payload_sensor	degradation / responsivity_decay	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-PAYLOAD-SENSOR-RESPONSIVITY-DECAY --output reports//CMP-DEG-PAYLOAD-SENSOR-RESPONSIVITY-DECAY.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-DEG-PAYLOAD-SENSOR-NOISE-GROWTH	component	payload_sensor	degradation / noise_growth	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-PAYLOAD-SENSOR-NOISE-GROWTH --output reports//CMP-DEG-PAYLOAD-SENSOR-NOISE-GROWTH.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				ng_verification_matrix.py --test-id CMP-DEG-PAYLOAD-SENSOR-NOISE-GROWTH --output reports//CMP-DEG-PAYLOAD-SENSOR-NOISE-GROWTH.json		
CMP-L2-PAYLOAD-SENSOR-FAULT-DEFAULT-SET	component	payload_sensor	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-PAYLOAD-SENSOR-FAULT-DEFAULT-SET --output reports//CMP-L2-PAYLOAD-SENSOR-FAULT-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.
CMP-L2-PAYLOAD-SENSOR-DEGRADATION-DEFAULT-SET	component	payload_sensor	degradation_default_set / degradation_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-PAYLOAD-SENSOR-DEGRADATION-DEFAULT-SET --output reports//CMP-L2-PAYLOAD-SENSOR-DEGRADATION-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default degradations participate at least once.
CMP-L2-PAYLOAD-SENSOR-COMBINED-DEFAULT-SET	component	payload_sensor	combined_default_set / combined_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-PAYLOAD-SENSOR-COMBINED-DEFAULT-SET --output reports//CMP-L2-PAYLOAD-SENSOR-COMBINED-	{"duration_s":600,"finite_numeric_required":true}	At least one fault and one degradation participate in one run.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				DEFAULT-SET.json		
CMP-BND-PAYLOAD-SENSOR-SHORT-DURATION-60S	component	payload_sensor	duration_boundary / short_duration_60s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-PAYLOAD-SENSOR-SHORT-DURATION-60S --output reports//CMP-BND-PAYLOAD-SENSOR-SHORT-DURATION-60S.json	{"duration_s":60.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-BND-PAYLOAD-SENSOR-LONG-DURATION-86400S	component	payload_sensor	duration_boundary / long_duration_86400s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-PAYLOAD-SENSOR-LONG-DURATION-86400S --output reports//CMP-BND-PAYLOAD-SENSOR-LONG-DURATION-86400S.json	{"duration_s":86400.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-NEG-PAYLOAD-SENSOR-UNSUPPORTED-WORKING-SCENARIO	component	payload_sensor	negative / unsupported_working_scenario	python scripts/run_engineering_verification_matrix.py --test-id CMP-NEG-PAYLOAD-SENSOR-UNSUPPORTED-WORKING-SCENARIO --output reports//CMP-NEG-PAYLOAD-SENSOR-UNSUPPORTED-WORKING-SCENARIO.json	{"expected_exception":"ValueError"}	Unsupported working scenario is rejected by the component runner.
CMP-DET-PAYLOAD-SENSOR-REPEAT-STABILITY	component	payload_sensor	determinism / repeat_stability	python scripts/run_engineering_verification_matrix.py	{"repeat_count":2,"semantic_hash_equal":true}	Two same-input runs produce the same semantic hash.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				x.py --test-id CMP-DET-PAYLOAD-SENSOR-REPEAT-STABILITY --output reports//CMP-DET-PAYLOAD-SENSOR-REPEAT-STABILITY.json		
CMP-NOM-PDU-NOMINAL	component	pdu	nominal / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-NOM-PDU-NOMINAL --output reports//CMP-NOM-PDU-NOMINAL.json	{"finite_numeric_required":true,"error_key_forbidden":true}	run_nominal_case returns structured data without error and all numeric outputs are finite.
CMP-WRK-PDU-NOMINAL	component	pdu	normal_working / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-PDU-NOMINAL --output reports//CMP-WRK-PDU-NOMINAL.json	{"finite_numeric_required":true,"scenario":"nominal"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-PDU-SAFE-MODE-LOADS	component	pdu	normal_working / safe_mode_loads	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-PDU-SAFE-MODE-LOADS --output reports//CMP-WRK-PDU-SAFE-MODE-LOADS.json	{"finite_numeric_required":true,"scenario":"safe_mode_loads"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-PDU-PAYLOAD-PEAK-LOADS	component	pdu	normal_working / payload_peak_loads	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-PDU-PAYLOAD-PEAK-LOADS --output reports//CMP-WRK-PDU-PAYLOAD-	{"finite_numeric_required":true,"scenario":"payload_peak_loads"}	Named normal operating scenario executes and returns finite outputs with no error field.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				PEAK-LOADS.json		
CMP-WRK-PDU-RECOVERY-LOADS	component	pdu	normal_working / recovery_loads	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-PDU-RECOVERY-LOADS --output reports//CMP-WRK-PDU-RECOVERY-LOADS.json	{"finite_numeric_required":true,"scenario":"recovery_loads"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-ALL-PDU-RUN-ALL-MODES	component	pdu	batch / run_all_modes	python scripts/run_engineering_verification_matrix.py --test-id CMP-ALL-PDU-RUN-ALL-MODES --output reports//CMP-ALL-PDU-RUN-ALL-MODES.json	{"minimum_top_level_modes":3,"finite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2-light coverage and finite numeric outputs.
CMP-FLT-PDU-CHANNEL-TRIP	component	pdu	fault / channel_trip	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-PDU-CHANNEL-TRIP --output reports//CMP-FLT-PDU-CHANNEL-TRIP.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-FLT-PDU-OVER-CURRENT	component	pdu	fault / over_current	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-PDU-OVER-CURRENT --output reports//CMP-FLT-PDU-OVER-CURRENT.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-DEG-PDU-CONTACT-RESISTANCE-GROWTH	component	pdu	degradation / contact_resistance_growth	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-PDU-CONTACT-RESISTANCE-GROWTH --output reports//CMP-DEG-PDU-CONTACT-RESISTANCE-GROWTH.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				x.py --test-id CMP-DEG-PDU-CONTACT-RESISTANCE-GROWTH --output reports//CMP-DEG-PDU-CONTACT-RESISTANCE-GROWTH.json		
CMP-DEG-PDU-VOLTAGE-REGULATION-DRIFT	component	pdu	degradation / voltage_regulation_drift	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-PDU-VOLTAGE-REGULATION-DRIFT --output reports//CMP-DEG-PDU-VOLTAGE-REGULATION-DRIFT.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-L2-PDU-FAULT-DEFAULT-SET	component	pdu	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-PDU-FAULT-DEFAULT-SET --output reports//CMP-L2-PDU-FAULT-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.
CMP-L2-PDU-DEGRADATION-DEFAULT-SET	component	pdu	degradation_default_set / degradation_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-PDU-DEGRADATION-DEFAULT-SET --output reports//CMP-L2-PDU-DEGRADATION-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default degradations participate at least once.
CMP-L2-PDU-COMBINED-DEFAULT-SET	component	pdu	combined_default_set / combined_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-PDU-COMBINED-DEFAULT-SET --output reports//CMP-L2-PDU-COMBINED-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	At least one fault and one degradation participate in one run.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				x.py --test-id CMP-L2-PDU-COMBINED-DEFAULT-SET --output reports//CMP-L2-PDU-COMBINED-DEFAULT-SET.json		
CMP-BND-PDU-SHORT-DURATION-60S	component	pdu	duration_boundary / short_duration_60s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-PDU-SHORT-DURATION-60S --output reports//CMP-BND-PDU-SHORT-DURATION-60S.json	{"duration_s":60.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-BND-PDU-LONG-DURATION-86400S	component	pdu	duration_boundary / long_duration_86400s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-PDU-LONG-DURATION-86400S --output reports//CMP-BND-PDU-LONG-DURATION-86400S.json	{"duration_s":86400.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-NEG-PDU-UNSUPPORTED-WORKING-SCENARIO	component	pdu	negative / unsupported_working_scenario	python scripts/run_engineering_verification_matrix.py --test-id CMP-NEG-PDU-UNSUPPORTED-WORKING-SCENARIO --output reports//CMP-NEG-PDU-UNSUPPORTED-WORKING-SCENARIO.json	{"expected_exception":"ValueError"}	Unsupported working scenario is rejected by the component runner.
CMP-DET-PDU-REPEAT-STABILITY	component	pdu	determinism / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id CMP-	{"repeat_count":2,"semantic_hash_equal":true}	Two same-input runs produce the same semantic hash.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				DET-PDU-REPEAT-STABILITY --output reports//CMP-DET-PDU-REPEAT-STABILITY.json		
CMP-NOM-POWER-SINK-NOMINAL	component	power_sink	nominal / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-NOM-POWER-SINK-NOMINAL --output reports//CMP-NOM-POWER-SINK-NOMINAL.json	{"finite_numeric_required":true,"error_key_forbidden":true}	run_nominal_case returns structured data without error and all numeric outputs are finite.
CMP-WRK-POWER-SINK-NOMINAL	component	power_sink	normal_working / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-POWER-SINK-NOMINAL --output reports//CMP-WRK-POWER-SINK-NOMINAL.json	{"finite_numeric_required":true,"scenario":"nominal"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-POWER-SINK-SAFE-MODE	component	power_sink	normal_working / safe_mode	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-POWER-SINK-SAFE-MODE --output reports//CMP-WRK-POWER-SINK-SAFE-MODE.json	{"finite_numeric_required":true,"scenario":"safe_mode"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-POWER-SINK-NOMINAL-LOAD	component	power_sink	normal_working / nominal_load	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-POWER-SINK-NOMINAL-LOAD --output reports//CMP-WRK-POWER-SINK-NOMINAL-LOAD.json	{"finite_numeric_required":true,"scenario":"nominal_load"}	Named normal operating scenario executes and returns finite outputs with no error field.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
CMP-WRK-POWER-SINK-PAYLOAD-PEAK	component	power_sink	normal_working / payload_peak	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-POWER-SINK-PAYLOAD-PEAK --output reports//CMP-WRK-POWER-SINK-PAYLOAD-PEAK.json	{"finite_numeric_required":true,"scenario":"payload_peak"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-ALL-POWER-SINK-RUN-ALL-MODES	component	power_sink	batch / run_all_modes	python scripts/run_engineering_verification_matrix.py --test-id CMP-ALL-POWER-SINK-RUN-ALL-MODES --output reports//CMP-ALL-POWER-SINK-RUN-ALL-MODES.json	{"minimum_top_level_modes":3,"finite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2-light coverage and finite numeric outputs.
CMP-FLT-POWER-SINK-OVERLOAD	component	power_sink	fault / overload	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-POWER-SINK-OVERLOAD --output reports//CMP-FLT-POWER-SINK-OVERLOAD.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-FLT-POWER-SINK-INTERMITTENT-LOAD	component	power_sink	fault / intermittent_load	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-POWER-SINK-INTERMITTENT-LOAD --output reports//CMP-FLT-POWER-SINK-INTERMITTENT-LOAD.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-DEG-POWER-SINK-LOAD-DRIFT	component	power_sink	degradation / load_drift	python scripts/run_engineeri	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				ng_verification_matrix.py --test-id CMP-DEG-POWER-SINK-LOAD-DRIFT --output reports//CMP-DEG-POWER-SINK-LOAD-DRIFT.json		
CMP-DEG-POWER-SINK-EFFICIENCY-DECAY	component	power_sink	degradation / efficiency_decay	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-POWER-SINK-EFFICIENCY-DECAY --output reports//CMP-DEG-POWER-SINK-EFFICIENCY-DECAY.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-L2-POWER-SINK-FAULT-DEFAULT-SET	component	power_sink	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-POWER-SINK-FAULT-DEFAULT-SET --output reports//CMP-L2-POWER-SINK-FAULT-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.
CMP-L2-POWER-SINK-DEGRADATION-DEFAULT-SET	component	power_sink	degradation_default_set / degradation_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-POWER-SINK-DEGRADATION-DEFAULT-SET --output reports//CMP-L2-POWER-SINK-DEGRADATION-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default degradations participate at least once.
CMP-L2-POWER-SINK-COMBINED-DEFAULT-SET	component	power_sink	combined_default_set / combined_default_set	python scripts/run_engineering_verification_matrix.py	{"duration_s":600,"finite_numeric_required":true}	At least one fault and one degradation participate in one run.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				x.py --test-id CMP-L2-POWER-SINK-COMBINED-DEFAULT-SET --output reports//CMP-L2-POWER-SINK-COMBINED-DEFAULT-SET.json		
CMP-BND-POWER-SINK-SHORT-DURATION-60S	component	power_sink	duration_boundary / short_duration_60s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-POWER-SINK-SHORT-DURATION-60S --output reports//CMP-BND-POWER-SINK-SHORT-DURATION-60S.json	{"duration_s":60.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-BND-POWER-SINK-LONG-DURATION-86400S	component	power_sink	duration_boundary / long_duration_86400s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-POWER-SINK-LONG-DURATION-86400S --output reports//CMP-BND-POWER-SINK-LONG-DURATION-86400S.json	{"duration_s":86400.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-NEG-POWER-SINK-UNSUPPORTED-WORKING-SCENARIO	component	power_sink	negative / unsupported_working_scenario	python scripts/run_engineering_verification_matrix.py --test-id CMP-NEG-POWER-SINK-UNSUPPORTED-WORKING-SCENARIO --output reports//CMP-NEG-POWER-SINK-	{"expected_exception":"ValueError"}	Unsupported working scenario is rejected by the component runner.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				UNSUPPORTED-WORKING-SCENARIO.json		
CMP-DET-POWER-SINK-REPEAT-STABILITY	component	power_sink	determinism / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id CMP-DET-POWER-SINK-REPEAT-STABILITY --output reports//CMP-DET-POWER-SINK-REPEAT-STABILITY.json	{"repeat_count":2,"semantic_hash_equal":true}	Two same-input runs produce the same semantic hash.
CMP-NOM-RADIATOR-NOMINAL	component	radiator	nominal / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-NOM-RADIATOR-NOMINAL --output reports//CMP-NOM-RADIATOR-NOMINAL.json	{"finite_numeric_required":true,"error_key_forbidden":true}	run_nominal_case returns structured data without error and all numeric outputs are finite.
CMP-ALL-RADIATOR-RUN-ALL-MODES	component	radiator	batch / run_all_modes	python scripts/run_engineering_verification_matrix.py --test-id CMP-ALL-RADIATOR-RUN-ALL-MODES --output reports//CMP-ALL-RADIATOR-RUN-ALL-MODES.json	{"minimum_top_level_modes":3,"finite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2-light coverage and finite numeric outputs.
CMP-FLT-RADIATOR-SURFACE-CONTAMINATION	component	radiator	fault / surface_contamination	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-RADIATOR-SURFACE-CONTAMINATION --output reports//CMP-FLT-RADIATOR-SURFACE-	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				CONTAMINATION.json		
CMP-FLT-RADIATOR-DEPLOY-FAILURE	component	radiator	fault / deploy_failure	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-RADIATOR-DEPLOY-FAILURE --output reports//CMP-FLT-RADIATOR-DEPLOY-FAILURE.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-DEG-RADIATOR-EMISSIVITY-DECAY	component	radiator	degradation / emissivity_decay	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-RADIATOR-EMISSIVITY-DECAY --output reports//CMP-DEG-RADIATOR-EMISSIVITY-DECAY.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-DEG-RADIATOR-ABSORPTIVITY-INCREASE	component	radiator	degradation / absorptivity_increase	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-RADIATOR-ABSORPTIVITY-INCREASE --output reports//CMP-DEG-RADIATOR-ABSORPTIVITY-INCREASE.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-L2-RADIATOR-FAULT-DEFAULT-SET	component	radiator	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-RADIATOR-FAULT-DEFAULT-SET --output reports//CMP-L2-RADIATOR-	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				FAULT-DEFAULT-SET.json		
CMP-L2-RADIATOR-DEGRADATION-DEFAULT-SET	component	radiator	degradation_default_set / degradation_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-RADIATOR-DEGRADATION-DEFAULT-SET --output reports//CMP-L2-RADIATOR-DEGRADATION-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default degradations participate at least once.
CMP-L2-RADIATOR-COMBINED-DEFAULT-SET	component	radiator	combined_default_set / combined_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-RADIATOR-COMBINED-DEFAULT-SET --output reports//CMP-L2-RADIATOR-COMBINED-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	At least one fault and one degradation participate in one run.
CMP-BND-RADIATOR-SHORT-DURATION-60S	component	radiator	duration_boundary / short_duration_60s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-RADIATOR-SHORT-DURATION-60S --output reports//CMP-BND-RADIATOR-SHORT-DURATION-60S.json	{"duration_s":60.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-BND-RADIATOR-LONG-DURATION-86400S	component	radiator	duration_boundary / long_duration_86400s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-RADIATOR-LONG-	{"duration_s":86400.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				DURATION-86400S --output reports//CMP-BND-RADIATOR-LONG-DURATION-86400S.json		
CMP-NEG-RADIATOR-UNSUPPORTED-WORKING-SCENARIO	component	radiator	negative / unsupported_working_scenario	python scripts/run_engineering_verification_matrix.py --test-id CMP-NEG-RADIATOR-UNSUPPORTED-WORKING-SCENARIO --output reports//CMP-NEG-RADIATOR-UNSUPPORTED-WORKING-SCENARIO.json	{"expected_exception":"ValueError"}	Unsupported working scenario is rejected by the component runner.
CMP-DET-RADIATOR-REPEAT-STABILITY	component	radiator	determinism / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id CMP-DET-RADIATOR-REPEAT-STABILITY --output reports//CMP-DET-RADIATOR-REPEAT-STABILITY.json	{"repeat_count":2,"semantic_hash_equal":true}	Two same-input runs produce the same semantic hash.
CMP-NOM-REACTION-WHEEL-NOMINAL	component	reaction_wheel	nominal / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-NOM-REACTION-WHEEL-NOMINAL --output reports//CMP-NOM-REACTION-WHEEL-NOMINAL.json	{"finite_numeric_required":true,"error_key_forbidden":true}	run_nominal_case returns structured data without error and all numeric outputs are finite.
CMP-ALL-REACTION-WHEEL-RUN-ALL-MODES	component	reaction_wheel	batch / run_all_modes	python scripts/run_engineeri	{"minimum_top_level_modes":3,"finite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2-light coverage and finite numeric

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				ng_verification_matrix.py --test-id CMP-ALL-REACTION-WHEEL-RUN-ALL-MODES --output reports//CMP-ALL-REACTION-WHEEL-RUN-ALL-MODES.json		outputs.
CMP-FLT-REACTION-WHEEL-BEARING-SEIZURE	component	reaction_wheel	fault / bearing_seizure	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-REACTION-WHEEL-BEARING-SEIZURE --output reports//CMP-FLT-REACTION-WHEEL-BEARING-SEIZURE.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-FLT-REACTION-WHEEL-MOTOR-OPEN	component	reaction_wheel	fault / motor_open	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-REACTION-WHEEL-MOTOR-OPEN --output reports//CMP-FLT-REACTION-WHEEL-MOTOR-OPEN.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-DEG-REACTION-WHEEL-BEARING-FRICTION-GROWTH	component	reaction_wheel	degradation / bearing_friction_growth	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-REACTION-WHEEL-BEARING-FRICTION-GROWTH --output reports//CMP-DEG-REACTION-WHEEL-BEARING-FRICTION-	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				GROWTH.json		
CMP-DEG-REACTION-WHEEL-TORQUE-CONSTANT-DRIFT	component	reaction_wheel	degradation / torque_constant_drift	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-REACTION-WHEEL-TORQUE-CONSTANT-DRIFT --output reports//CMP-DEG-REACTION-WHEEL-TORQUE-CONSTANT-DRIFT.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-L2-REACTION-WHEEL-FAULT-DEFAULT-SET	component	reaction_wheel	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-REACTION-WHEEL-FAULT-DEFAULT-SET --output reports//CMP-L2-REACTION-WHEEL-FAULT-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.
CMP-L2-REACTION-WHEEL-DEGRADATION-DEFAULT-SET	component	reaction_wheel	degradation_default_set / degradation_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-REACTION-WHEEL-DEGRADATION-DEFAULT-SET --output reports//CMP-L2-REACTION-WHEEL-DEGRADATION-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default degradations participate at least once.
CMP-L2-REACTION-WHEEL-COMBINED-DEFAULT-SET	component	reaction_wheel	combined_default_set / combined_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-REACTION-WHEEL-	{"duration_s":600,"finite_numeric_required":true}	At least one fault and one degradation participate in one run.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				COMBINED- DEFAULT-SET -- output reports//CMP- L2-REACTION- WHEEL-COMBINED- DEFAULT-SET.json		
CMP-BND-REACTION-WHEEL- SHORT-DURATION-60S	component	reaction_wheel	duration_boundary / short_duration_60s	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- BND-REACTION- WHEEL-SHORT- DURATION-60S -- output reports//CMP- BND-REACTION- WHEEL-SHORT- DURATION-60S.json	{"duration_s":60.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-BND-REACTION-WHEEL- LONG-DURATION-86400S	component	reaction_wheel	duration_boundary / long_duration_86400s	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- BND-REACTION- WHEEL-LONG- DURATION-86400S -- output reports//CMP- BND-REACTION- WHEEL-LONG- DURATION-86400S.js on	{"duration_s":86400.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-NEG-REACTION-WHEEL- UNSUPPORTED-WORKING- SCENARIO	component	reaction_wheel	negative / unsupported_working_scenario	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- NEG-REACTION- WHEEL- UNSUPPORTED- WORKING- SCENARIO --output reports//CMP-NEG- REACTION-WHEEL- UNSUPPORTED-	{"expected_exception":"ValueError "}	Unsupported working scenario is rejected by the component runner.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				WORKING-SCENARIO.json		
CMP-DET-REACTION-WHEEL-REPEAT-STABILITY	component	reaction_wheel	determinism / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id CMP-DET-REACTION-WHEEL-REPEAT-STABILITY --output reports//CMP-DET-REACTION-WHEEL-REPEAT-STABILITY.json	{"repeat_count":2,"semantic_hash_equal":true}	Two same-input runs produce the same semantic hash.
CMP-NOM-SOLAR-PANEL-NOMINAL	component	solar_panel	nominal / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-NOM-SOLAR-PANEL-NOMINAL --output reports//CMP-NOM-SOLAR-PANEL-NOMINAL.json	{"finite_numeric_required":true,"error_key_forbidden":true}	run_nominal_case returns structured data without error and all numeric outputs are finite.
CMP-ALL-SOLAR-PANEL-RUN-ALL-MODES	component	solar_panel	batch / run_all_modes	python scripts/run_engineering_verification_matrix.py --test-id CMP-ALL-SOLAR-PANEL-RUN-ALL-MODES --output reports//CMP-ALL-SOLAR-PANEL-RUN-ALL-MODES.json	{"minimum_top_level_modes":3,"finite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2-light coverage and finite numeric outputs.
CMP-FLT-SOLAR-PANEL-STRING-OPEN	component	solar_panel	fault / string_open	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-SOLAR-PANEL-STRING-OPEN --output reports//CMP-FLT-SOLAR-PANEL-STRING-OPEN.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
CMP-FLT-SOLAR-PANEL-SHADING	component	solar_panel	fault / shading	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-SOLAR-PANEL-SHADING --output reports//CMP-FLT-SOLAR-PANEL-SHADING.json	{"active_fault_required":true,"sample_window_s":{0,600}}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-DEG-SOLAR-PANEL-RADIATION-EFFICIENCY-LOSS	component	solar_panel	degradation / radiation_efficiency_loss	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-SOLAR-PANEL-RADIATION-EFFICIENCY-LOSS --output reports//CMP-DEG-SOLAR-PANEL-RADIATION-EFFICIENCY-LOSS.json	{"active_degradation_required":true,"sample_window_s":{0,600}}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-DEG-SOLAR-PANEL-COVER-GLASS-DARKENING	component	solar_panel	degradation / cover_glass_darkening	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-SOLAR-PANEL-COVER-GLASS-DARKENING --output reports//CMP-DEG-SOLAR-PANEL-COVER-GLASS-DARKENING.json	{"active_degradation_required":true,"sample_window_s":{0,600}}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-L2-SOLAR-PANEL-FAULT-DEFAULT-SET	component	solar_panel	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-SOLAR-PANEL-FAULT-DEFAULT-SET --output reports//CMP-L2-SOLAR-PANEL-	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				FAULT-DEFAULT-SET.json		
CMP-L2-SOLAR-PANEL-DEGRADATION-DEFAULT-SET	component	solar_panel	degradation_default_set / degradation_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-SOLAR-PANEL-DEGRADATION-DEFAULT-SET --output reports//CMP-L2-SOLAR-PANEL-DEGRADATION-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default degradations participate at least once.
CMP-L2-SOLAR-PANEL-COMBINED-DEFAULT-SET	component	solar_panel	combined_default_set / combined_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-SOLAR-PANEL-COMBINED-DEFAULT-SET --output reports//CMP-L2-SOLAR-PANEL-COMBINED-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	At least one fault and one degradation participate in one run.
CMP-BND-SOLAR-PANEL-SHORT-DURATION-60S	component	solar_panel	duration_boundary / short_duration_60s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-SOLAR-PANEL-SHORT-DURATION-60S --output reports//CMP-BND-SOLAR-PANEL-SHORT-DURATION-60S.json	{"duration_s":60.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-BND-SOLAR-PANEL-LONG-DURATION-86400S	component	solar_panel	duration_boundary / long_duration_86400s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-SOLAR-PANEL-LONG-	{"duration_s":86400.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				DURATION-86400S --output reports//CMP-BND-SOLAR-PANEL-LONG-DURATION-86400S.json		
CMP-NEG-SOLAR-PANEL-UNSUPPORTED-WORKING-SCENARIO	component	solar_panel	negative / unsupported_working_scenario	python scripts/run_engineering_verification_matrix.py --test-id CMP-NEG-SOLAR-PANEL-UNSUPPORTED-WORKING-SCENARIO --output reports//CMP-NEG-SOLAR-PANEL-UNSUPPORTED-WORKING-SCENARIO.json	{"expected_exception":"ValueError"}	Unsupported working scenario is rejected by the component runner.
CMP-DET-SOLAR-PANEL-REPEAT-STABILITY	component	solar_panel	determinism / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id CMP-DET-SOLAR-PANEL-REPEAT-STABILITY --output reports//CMP-DET-SOLAR-PANEL-REPEAT-STABILITY.json	{"repeat_count":2,"semantic_hash_equal":true}	Two same-input runs produce the same semantic hash.
CMP-NOM-STAR-TRACKER-NOMINAL	component	star_tracker	nominal / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-NOM-STAR-TRACKER-NOMINAL --output reports//CMP-NOM-STAR-TRACKER-NOMINAL.json	{"finite_numeric_required":true,"error_key_forbidden":true}	run_nominal_case returns structured data without error and all numeric outputs are finite.
CMP-WRK-STAR-TRACKER-NOMINAL	component	star_tracker	normal_working / nominal	python scripts/run_engineeri	{"finite_numeric_required":true,"scenario":"nominal"}	Named normal operating scenario executes and returns finite outputs with no error field.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				ng_verification_matrix.py --test-id CMP-WRK-STAR-TRACKER-NOMINAL --output reports//CMP-WRK-STAR-TRACKER-NOMINAL.json		
CMP-WRK-STAR-TRACKER-STABLE-ATTITUDE	component	star_tracker	normal_working / stable_attitude	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-STAR-TRACKER-STABLE-ATTITUDE --output reports//CMP-WRK-STAR-TRACKER-STABLE-ATTITUDE.json	{"finite_numeric_required":true,"scenario":"stable_attitude"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-STAR-TRACKER-SLEW-PROFILE	component	star_tracker	normal_working / slew_profile	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-STAR-TRACKER-SLEW-PROFILE --output reports//CMP-WRK-STAR-TRACKER-SLEW-PROFILE.json	{"finite_numeric_required":true,"scenario":"slew_profile"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-STAR-TRACKER-LONG-DRIFT	component	star_tracker	normal_working / long_drift	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-STAR-TRACKER-LONG-DRIFT --output reports//CMP-WRK-STAR-TRACKER-LONG-DRIFT.json	{"finite_numeric_required":true,"scenario":"long_drift"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-ALL-STAR-TRACKER-RUN-ALL-MODES	component	star_tracker	batch / run_all_modes	python scripts/run_engineeri	{"minimum_top_level_modes":3,"finite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2-light coverage and finite numeric

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				ng_verification_matri x.py --test-id CMP- ALL-STAR-TRACKER- RUN-ALL-MODES -- output reports//CMP- ALL-STAR-TRACKER- RUN-ALL- MODES.json		outputs.
CMP-FLT-STAR-TRACKER- BLINDING	component	star_tracker	fault / blinding	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- FLT-STAR-TRACKER- BLINDING --output reports//CMP-FLT- STAR-TRACKER- BLINDING.json	{"active_fault_required":true,"sam ple_window_s":{0,600}}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-FLT-STAR-TRACKER- DROPOUT	component	star_tracker	fault / dropout	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- FLT-STAR-TRACKER- DROPOUT --output reports//CMP-FLT- STAR-TRACKER- DROPOUT.json	{"active_fault_required":true,"sam ple_window_s":{0,600}}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-DEG-STAR-TRACKER- CENTROID-NOISE-GROWTH	component	star_tracker	degradation / centroid_noise_growth	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- DEG-STAR-TRACKER- CENTROID-NOISE- GROWTH --output reports//CMP-DEG- STAR-TRACKER- CENTROID-NOISE- GROWTH.json	{"active_degradation_required":tru e,"sample_window_s":{0,600}}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-DEG-STAR-TRACKER- BORESIGHT-DRIFT	component	star_tracker	degradation / boresight_drift	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP-	{"active_degradation_required":tru e,"sample_window_s":{0,600}}	The single degradation is active after onset and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				DEG-STAR-TRACKER-BORESIGHT-DRIFT --output reports//CMP-DEG-STAR-TRACKER-BORESIGHT-DRIFT.json		
CMP-L2-STAR-TRACKER-FAULT-DEFAULT-SET	component	star_tracker	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-STAR-TRACKER-FAULT-DEFAULT-SET --output reports//CMP-L2-STAR-TRACKER-FAULT-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.
CMP-L2-STAR-TRACKER-DEGRADATION-DEFAULT-SET	component	star_tracker	degradation_default_set / degradation_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-STAR-TRACKER-DEGRADATION-DEFAULT-SET --output reports//CMP-L2-STAR-TRACKER-DEGRADATION-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default degradations participate at least once.
CMP-L2-STAR-TRACKER-COMBINED-DEFAULT-SET	component	star_tracker	combined_default_set / combined_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-STAR-TRACKER-COMBINED-DEFAULT-SET --output reports//CMP-L2-STAR-TRACKER-COMBINED-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	At least one fault and one degradation participate in one run.
CMP-BND-STAR-TRACKER-SHORT-DURATION-60S	component	star_tracker	duration_boundary / short_duration_60s	python scripts/run_engineeri	{"duration_s":60.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				ng_verification_matrix.py --test-id CMP-BND-STAR-TRACKER-SHORT-DURATION-60S --output reports//CMP-BND-STAR-TRACKER-SHORT-DURATION-60S.json		count.
CMP-BND-STAR-TRACKER-LONG-DURATION-86400S	component	star_tracker	duration_boundary / long_duration_86400s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-STAR-TRACKER-LONG-DURATION-86400S --output reports//CMP-BND-STAR-TRACKER-LONG-DURATION-86400S.json	{"duration_s":86400.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-NEG-STAR-TRACKER-UNSUPPORTED-WORKING-SCENARIO	component	star_tracker	negative / unsupported_working_scenario	python scripts/run_engineering_verification_matrix.py --test-id CMP-NEG-STAR-TRACKER-UNSUPPORTED-WORKING-SCENARIO --output reports//CMP-NEG-STAR-TRACKER-UNSUPPORTED-WORKING-SCENARIO.json	{"expected_exception":"ValueError"}	Unsupported working scenario is rejected by the component runner.
CMP-DET-STAR-TRACKER-REPEAT-STABILITY	component	star_tracker	determinism / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id CMP-DET-STAR-TRACKER-REPEAT-STABILITY --output reports//CMP-	{"repeat_count":2,"semantic_hash_equal":true}	Two same-input runs produce the same semantic hash.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				DET-STAR-TRACKER-REPEAT-STABILITY.json		
CMP-NOM-SUN-SENSOR-NOMINAL	component	sun_sensor	nominal / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-NOM-SUN-SENSOR-NOMINAL --output reports//CMP-NOM-SUN-SENSOR-NOMINAL.json	{"finite_numeric_required":true,"error_key_forbidden":true}	run_nominal_case returns structured data without error and all numeric outputs are finite.
CMP-WRK-SUN-SENSOR-NOMINAL	component	sun_sensor	normal_working / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-SUN-SENSOR-NOMINAL --output reports//CMP-WRK-SUN-SENSOR-NOMINAL.json	{"finite_numeric_required":true,"scenario":"nominal"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-SUN-SENSOR-FULL-SUN	component	sun_sensor	normal_working / full_sun	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-SUN-SENSOR-FULL-SUN --output reports//CMP-WRK-SUN-SENSOR-FULL-SUN.json	{"finite_numeric_required":true,"scenario":"full_sun"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-SUN-SENSOR-PARTIAL-SHADOW	component	sun_sensor	normal_working / partial_shadow	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-SUN-SENSOR-PARTIAL-SHADOW --output reports//CMP-WRK-SUN-SENSOR-PARTIAL-SHADOW.json	{"finite_numeric_required":true,"scenario":"partial_shadow"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-SUN-SENSOR-ECLIPSE	component	sun_sensor	normal_working / eclipse	python	{"finite_numeric_required":true,"sc	Named normal operating scenario

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-SUN-SENSOR-ECLIPSE --output reports//CMP-WRK-SUN-SENSOR-ECLIPSE.json	enario":"eclipse"}	executes and returns finite outputs with no error field.
CMP-ALL-SUN-SENSOR-RUN-ALL-MODES	component	sun_sensor	batch / run_all_modes	python scripts/run_engineering_verification_matrix.py --test-id CMP-ALL-SUN-SENSOR-RUN-ALL-MODES --output reports//CMP-ALL-SUN-SENSOR-RUN-ALL-MODES.json	{"minimum_top_level_modes":3,"finite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2-light coverage and finite numeric outputs.
CMP-FLT-SUN-SENSOR-FALSE-ECLIPSE	component	sun_sensor	fault / false_eclipse	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-SUN-SENSOR-FALSE-ECLIPSE --output reports//CMP-FLT-SUN-SENSOR-FALSE-ECLIPSE.json	{"active_fault_required":true,"sample_window_s":{0,600}}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-FLT-SUN-SENSOR-CELL-FAILURE	component	sun_sensor	fault / cell_failure	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-SUN-SENSOR-CELL-FAILURE --output reports//CMP-FLT-SUN-SENSOR-CELL-FAILURE.json	{"active_fault_required":true,"sample_window_s":{0,600}}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-DEG-SUN-SENSOR-SENSITIVITY-DECAY	component	sun_sensor	degradation / sensitivity_decay	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-SUN-SENSOR-	{"active_degradation_required":true,"sample_window_s":{0,600}}	The single degradation is active after onset and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				SENSITIVITY-DECAY --output reports//CMP-DEG- SUN-SENSOR- SENSITIVITY- DECAY.json		
CMP-DEG-SUN-SENSOR-BIAS- DRIFT	component	sun_sensor	degradation / bias_drift	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- DEG-SUN-SENSOR- BIAS-DRIFT --output reports//CMP-DEG- SUN-SENSOR-BIAS- DRIFT.json	{"active_degradation_required":tru e,"sample_window_s":{0,600}}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-L2-SUN-SENSOR-FAULT- DEFAULT-SET	component	sun_sensor	fault_default_set / fault_default_set	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP-L2- SUN-SENSOR-FAULT- DEFAULT-SET -- output reports//CMP- L2-SUN-SENSOR- FAULT-DEFAULT- SET.json	{"duration_s":600,"finite_numeric_ required":true}	All default faults participate at least once.
CMP-L2-SUN-SENSOR- DEGRADATION-DEFAULT-SET	component	sun_sensor	degradation_default_set / degradation_default_set	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP-L2- SUN-SENSOR- DEGRADATION- DEFAULT-SET -- output reports//CMP- L2-SUN-SENSOR- DEGRADATION- DEFAULT-SET.json	{"duration_s":600,"finite_numeric_ required":true}	All default degradations participate at least once.
CMP-L2-SUN-SENSOR-COMBINED- DEFAULT-SET	component	sun_sensor	combined_default_set / combined_default_set	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP-L2- SUN-SENSOR-	{"duration_s":600,"finite_numeric_ required":true}	At least one fault and one degradation participate in one run.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				COMBINED- DEFAULT-SET -- output reports//CMP- L2-SUN-SENSOR- COMBINED- DEFAULT-SET.json		
CMP-BND-SUN-SENSOR-SHORT- DURATION-60S	component	sun_sensor	duration_boundary / short_duration_60s	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- BND-SUN-SENSOR- SHORT- DURATION-60S -- output reports//CMP- BND-SUN-SENSOR- SHORT- DURATION-60S.json	{"duration_s":60.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-BND-SUN-SENSOR-LONG- DURATION-86400S	component	sun_sensor	duration_boundary / long_duration_86400s	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- BND-SUN-SENSOR- LONG- DURATION-86400S -- output reports//CMP- BND-SUN-SENSOR- LONG- DURATION-86400S.js on	{"duration_s":86400.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-NEG-SUN-SENSOR- UNSUPPORTED-WORKING- SCENARIO	component	sun_sensor	negative / unsupported_working_scenario	python scripts/run_engineeri ng_verification_matri x.py --test-id CMP- NEG-SUN-SENSOR- UNSUPPORTED- WORKING- SCENARIO --output reports//CMP-NEG- SUN-SENSOR- UNSUPPORTED- WORKING-	{"expected_exception":"ValueError "}	Unsupported working scenario is rejected by the component runner.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				SCENARIO.json		
CMP-DET-SUN-SENSOR-REPEAT-STABILITY	component	sun_sensor	determinism / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id CMP-DET-SUN-SENSOR-REPEAT-STABILITY --output reports//CMP-DET-SUN-SENSOR-REPEAT-STABILITY.json	{"repeat_count":2,"semantic_hash_equal":true}	Two same-input runs produce the same semantic hash.
CMP-NOM-THERMAL-NODE-NOMINAL	component	thermal_node	nominal / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-NOM-THERMAL-NODE-NOMINAL --output reports//CMP-NOM-THERMAL-NODE-NOMINAL.json	{"finite_numeric_required":true,"error_key_forbidden":true}	run_nominal_case returns structured data without error and all numeric outputs are finite.
CMP-WRK-THERMAL-NODE-NOMINAL	component	thermal_node	normal_working / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-THERMAL-NODE-NOMINAL --output reports//CMP-WRK-THERMAL-NODE-NOMINAL.json	{"finite_numeric_required":true,"scenario":"nominal"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-THERMAL-NODE-COLD-SOAK-RECOVERY	component	thermal_node	normal_working / cold_soak_recovery	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-THERMAL-NODE-COLD-SOAK-RECOVERY --output reports//CMP-WRK-THERMAL-NODE-COLD-SOAK-RECOVERY.json	{"finite_numeric_required":true,"scenario":"cold_soak_recovery"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-THERMAL-NODE-	component	thermal_node	normal_working / payload_heating	python	{"finite_numeric_required":true,"sc	Named normal operating scenario

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
PAYLOAD-HEATING				scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-THERMAL-NODE-PAYLOAD-HEATING --output reports//CMP-WRK-THERMAL-NODE-PAYLOAD-HEATING.json	enario:"payload_heating"}	executes and returns finite outputs with no error field.
CMP-WRK-THERMAL-NODE-COOLDOWN	component	thermal_node	normal_working / cooldown	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-THERMAL-NODE-COOLDOWN --output reports//CMP-WRK-THERMAL-NODE-COOLDOWN.json	{"finite_numeric_required":true,"scenario":"cooldown"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-ALL-THERMAL-NODE-RUN-ALL-MODES	component	thermal_node	batch / run_all_modes	python scripts/run_engineering_verification_matrix.py --test-id CMP-ALL-THERMAL-NODE-RUN-ALL-MODES --output reports//CMP-ALL-THERMAL-NODE-RUN-ALL-MODES.json	{"minimum_top_level_modes":3,"finite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2-light coverage and finite numeric outputs.
CMP-FLT-THERMAL-NODE-THERMAL-RUNAWAY	component	thermal_node	fault / thermal_runaway	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-THERMAL-NODE-THERMAL-RUNAWAY --output reports//CMP-FLT-THERMAL-NODE-THERMAL-	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				RUNAWAY.json		
CMP-FLT-THERMAL-NODE-SENSOR-BIAS	component	thermal_node	fault / sensor_bias	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-THERMAL-NODE-SENSOR-BIAS --output reports//CMP-FLT-THERMAL-NODE-SENSOR-BIAS.json	{"active_fault_required":true,"sample_window_s":{0,600}}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-DEG-THERMAL-NODE-THERMAL-RESISTANCE-GROWTH	component	thermal_node	degradation / thermal_resistance_growth	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-THERMAL-NODE-THERMAL-RESISTANCE-GROWTH --output reports//CMP-DEG-THERMAL-NODE-THERMAL-RESISTANCE-GROWTH.json	{"active_degradation_required":true,"sample_window_s":{0,600}}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-DEG-THERMAL-NODE-HEAT-CAPACITY-DRIFT	component	thermal_node	degradation / heat_capacity_drift	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-THERMAL-NODE-HEAT-CAPACITY-DRIFT --output reports//CMP-DEG-THERMAL-NODE-HEAT-CAPACITY-DRIFT.json	{"active_degradation_required":true,"sample_window_s":{0,600}}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-L2-THERMAL-NODE-FAULT-DEFAULT-SET	component	thermal_node	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-THERMAL-NODE-	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				FAULT-DEFAULT-SET --output reports//CMP-L2-THERMAL-NODE-FAULT-DEFAULT-SET.json		
CMP-L2-THERMAL-NODE-DEGRADATION-DEFAULT-SET	component	thermal_node	degradation_default_set / degradation_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-THERMAL-NODE-DEGRADATION-DEFAULT-SET --output reports//CMP-L2-THERMAL-NODE-DEGRADATION-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default degradations participate at least once.
CMP-L2-THERMAL-NODE-COMBINED-DEFAULT-SET	component	thermal_node	combined_default_set / combined_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-THERMAL-NODE-COMBINED-DEFAULT-SET --output reports//CMP-L2-THERMAL-NODE-COMBINED-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	At least one fault and one degradation participate in one run.
CMP-BND-THERMAL-NODE-SHORT-DURATION-60S	component	thermal_node	duration_boundary / short_duration_60s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-THERMAL-NODE-SHORT-DURATION-60S --output reports//CMP-BND-THERMAL-NODE-SHORT-DURATION-60S.json	{"duration_s":60.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-BND-THERMAL-NODE-LONG-DURATION-86400S	component	thermal_node	duration_boundary / long_duration_86400s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-THERMAL-NODE-LONG-DURATION-86400S --output reports//CMP-BND-THERMAL-NODE-LONG-DURATION-86400S.json	{"duration_s":86400.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				ng_verification_matrix.py --test-id CMP-BND-THERMAL-NODE-LONG-DURATION-86400S --output reports//CMP-BND-THERMAL-NODE-LONG-DURATION-86400S.json		count.
CMP-NEG-THERMAL-NODE-UNSUPPORTED-WORKING-SCENARIO	component	thermal_node	negative / unsupported_working_scenario	python scripts/run_engineering_verification_matrix.py --test-id CMP-NEG-THERMAL-NODE-UNSUPPORTED-WORKING-SCENARIO --output reports//CMP-NEG-THERMAL-NODE-UNSUPPORTED-WORKING-SCENARIO.json	{"expected_exception":"ValueError"}	Unsupported working scenario is rejected by the component runner.
CMP-DET-THERMAL-NODE-REPEAT-STABILITY	component	thermal_node	determinism / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id CMP-DET-THERMAL-NODE-REPEAT-STABILITY --output reports//CMP-DET-THERMAL-NODE-REPEAT-STABILITY.json	{"repeat_count":2,"semantic_hash_equal":true}	Two same-input runs produce the same semantic hash.
CMP-NOM-THRUSTER-NOMINAL	component	thruster	nominal / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-NOM-THRUSTER-NOMINAL --output	{"finite_numeric_required":true,"error_key_forbidden":true}	run_nominal_case returns structured data without error and all numeric outputs are finite.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				reports//CMP-NOM-THRUSTER-NOMINAL.json		
CMP-WRK-THRUSTER-NOMINAL	component	thruster	normal_working / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-THRUSTER-NOMINAL --output reports//CMP-WRK-THRUSTER-NOMINAL.json	{"finite_numeric_required":true,"scenario":"nominal"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-THRUSTER-SHORT-IMPULSE	component	thruster	normal_working / short_impulse	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-THRUSTER-SHORT-IMPULSE --output reports//CMP-WRK-THRUSTER-SHORT-IMPULSE.json	{"finite_numeric_required":true,"scenario":"short_impulse"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-THRUSTER-STATION-KEEPING	component	thruster	normal_working / station_keeping	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-THRUSTER-STATION-KEEPING --output reports//CMP-WRK-THRUSTER-STATION-KEEPING.json	{"finite_numeric_required":true,"scenario":"station_keeping"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-THRUSTER-LONG-BURN	component	thruster	normal_working / long_burn	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-THRUSTER-LONG-BURN --output reports//CMP-WRK-THRUSTER-LONG-BURN.json	{"finite_numeric_required":true,"scenario":"long_burn"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-ALL-THRUSTER-RUN-ALL-	component	thruster	batch / run_all_modes	python	{"minimum_top_level_modes":3,"fi	run_all_modes returns a dict with

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
MODES				scripts/run_engineering_verification_matrix.py --test-id CMP-ALL-THRUSTER-RUN-ALL-MODES --output reports//CMP-ALL-THRUSTER-RUN-ALL-MODES.json	nite_numeric_required":true}	nominal/fault/degradation or L2-light coverage and finite numeric outputs.
CMP-FLT-THRUSTER-VALVE-STUCK-CLOSED	component	thruster	fault / valve_stuck_closed	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-THRUSTER-VALVE-STUCK-CLOSED --output reports//CMP-FLT-THRUSTER-VALVE-STUCK-CLOSED.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-FLT-THRUSTER-VALVE-STUCK-OPEN	component	thruster	fault / valve_stuck_open	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-THRUSTER-VALVE-STUCK-OPEN --output reports//CMP-FLT-THRUSTER-VALVE-STUCK-OPEN.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-DEG-THRUSTER-ISP-DECAY	component	thruster	degradation / isp_decay	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-THRUSTER-ISP-DECAY --output reports//CMP-DEG-THRUSTER-ISP-DECAY.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-DEG-THRUSTER-THRUST-COEFFICIENT-DRIFT	component	thruster	degradation / thrust_coefficient_drift	python scripts/run_engineering_verification_matrix.py --test-id CMP-	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				DEG-THRUSTER-THRUST-COEFFICIENT-DRIFT --output reports//CMP-DEG-THRUSTER-THRUST-COEFFICIENT-DRIFT.json		
CMP-L2-THRUSTER-FAULT-DEFAULT-SET	component	thruster	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-THRUSTER-FAULT-DEFAULT-SET --output reports//CMP-L2-THRUSTER-FAULT-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.
CMP-L2-THRUSTER-DEGRADATION-DEFAULT-SET	component	thruster	degradation_default_set / degradation_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-THRUSTER-DEGRADATION-DEFAULT-SET --output reports//CMP-L2-THRUSTER-DEGRADATION-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default degradations participate at least once.
CMP-L2-THRUSTER-COMBINED-DEFAULT-SET	component	thruster	combined_default_set / combined_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-THRUSTER-COMBINED-DEFAULT-SET --output reports//CMP-L2-THRUSTER-COMBINED-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	At least one fault and one degradation participate in one run.
CMP-BND-THRUSTER-SHORT-	component	thruster	duration_boundary /	python	{"duration_s":60.0,"finite_numeric_	Boundary-duration combined

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
DURATION-60S			short_duration_60s	scripts/run_engineering_verification_matrix.py --test-id CMP-BND-THRUSTER-SHORT-DURATION-60S --output reports//CMP-BND-THRUSTER-SHORT-DURATION-60S.json	required":true}	sparse event run completes with finite outputs and bounded sample count.
CMP-BND-THRUSTER-LONG-DURATION-86400S	component	thruster	duration_boundary / long_duration_86400s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-THRUSTER-LONG-DURATION-86400S --output reports//CMP-BND-THRUSTER-LONG-DURATION-86400S.json	{"duration_s":86400.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-NEG-THRUSTER-UNSUPPORTED-WORKING-SCENARIO	component	thruster	negative / unsupported_working_scenario	python scripts/run_engineering_verification_matrix.py --test-id CMP-NEG-THRUSTER-UNSUPPORTED-WORKING-SCENARIO --output reports//CMP-NEG-THRUSTER-UNSUPPORTED-WORKING-SCENARIO.json	{"expected_exception":"ValueError"}	Unsupported working scenario is rejected by the component runner.
CMP-DET-THRUSTER-REPEAT-STABILITY	component	thruster	determinism / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id CMP-DET-THRUSTER-REPEAT-STABILITY --	{"repeat_count":2,"semantic_hash_equal":true}	Two same-input runs produce the same semantic hash.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				output reports//CMP-DET-THRUSTER-REPEAT-STABILITY.json		
CMP-NOM-TRANSMITTER-NOMINAL	component	transmitter	nominal / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-NOM-TRANSMITTER-NOMINAL --output reports//CMP-NOM-TRANSMITTER-NOMINAL.json	{"finite_numeric_required":true,"error_key_forbidden":true}	run_nominal_case returns structured data without error and all numeric outputs are finite.
CMP-WRK-TRANSMITTER-NOMINAL	component	transmitter	normal_working / nominal	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-TRANSMITTER-NOMINAL --output reports//CMP-WRK-TRANSMITTER-NOMINAL.json	{"finite_numeric_required":true,"scenario":"nominal"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-TRANSMITTER-TWT-DOWNLINK	component	transmitter	normal_working / twt_downlink	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-TRANSMITTER-TWT-DOWNLINK --output reports//CMP-WRK-TRANSMITTER-TWT-DOWNLINK.json	{"finite_numeric_required":true,"scenario":"twt_downlink"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-WRK-TRANSMITTER-SSPA-DOWNLINK	component	transmitter	normal_working / sspa_downlink	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-TRANSMITTER-SSPA-DOWNLINK --output reports//CMP-WRK-TRANSMITTER-SSPA-	{"finite_numeric_required":true,"scenario":"sspa_downlink"}	Named normal operating scenario executes and returns finite outputs with no error field.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				DOWNLINK.json		
CMP-WRK-TRANSMITTER-BURST-CONTACT	component	transmitter	normal_working / burst_contact	python scripts/run_engineering_verification_matrix.py --test-id CMP-WRK-TRANSMITTER-BURST-CONTACT --output reports//CMP-WRK-TRANSMITTER-BURST-CONTACT.json	{"finite_numeric_required":true,"scenario":"burst_contact"}	Named normal operating scenario executes and returns finite outputs with no error field.
CMP-ALL-TRANSMITTER-RUN-ALL-MODES	component	transmitter	batch / run_all_modes	python scripts/run_engineering_verification_matrix.py --test-id CMP-ALL-TRANSMITTER-RUN-ALL-MODES --output reports//CMP-ALL-TRANSMITTER-RUN-ALL-MODES.json	{"minimum_top_level_modes":3,"finite_numeric_required":true}	run_all_modes returns a dict with nominal/fault/degradation or L2-light coverage and finite numeric outputs.
CMP-FLT-TRANSMITTER-POWER-AMPLIFIER-FAULT	component	transmitter	fault / power_amplifier_fault	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-TRANSMITTER-POWER-AMPLIFIER-FAULT --output reports//CMP-FLT-TRANSMITTER-POWER-AMPLIFIER-FAULT.json	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.
CMP-FLT-TRANSMITTER-FREQUENCY-DRIFT-FAULT	component	transmitter	fault / frequency_drift_fault	python scripts/run_engineering_verification_matrix.py --test-id CMP-FLT-TRANSMITTER-FREQUENCY-DRIFT-FAULT --output reports//CMP-FLT-TRANSMITTER-	{"active_fault_required":true,"sample_window_s":[0,600]}	The single fault is active in its configured time window and changes effective parameters or output evidence.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				FREQUENCY-DRIFT-FAULT.json		
CMP-DEG-TRANSMITTER-OUTPUT-POWER-DECAY	component	transmitter	degradation / output_power_decay	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-TRANSMITTER-OUTPUT-POWER-DECAY --output reports//CMP-DEG-TRANSMITTER-OUTPUT-POWER-DECAY.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-DEG-TRANSMITTER-FREQUENCY-STABILITY-DRIFT	component	transmitter	degradation / frequency_stability_drift	python scripts/run_engineering_verification_matrix.py --test-id CMP-DEG-TRANSMITTER-FREQUENCY-STABILITY-DRIFT --output reports//CMP-DEG-TRANSMITTER-FREQUENCY-STABILITY-DRIFT.json	{"active_degradation_required":true,"sample_window_s":[0,600]}	The single degradation is active after onset and changes effective parameters or output evidence.
CMP-L2-TRANSMITTER-FAULT-DEFAULT-SET	component	transmitter	fault_default_set / fault_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-TRANSMITTER-FAULT-DEFAULT-SET --output reports//CMP-L2-TRANSMITTER-FAULT-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	All default faults participate at least once.
CMP-L2-TRANSMITTER-DEGRADATION-DEFAULT-SET	component	transmitter	degradation_default_set / degradation_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-TRANSMITTER-	{"duration_s":600,"finite_numeric_required":true}	All default degradations participate at least once.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				DEGRADATION-DEFAULT-SET --output reports//CMP-L2-TRANSMITTER-DEGRADATION-DEFAULT-SET.json		
CMP-L2-TRANSMITTER-COMBINED-DEFAULT-SET	component	transmitter	combined_default_set / combined_default_set	python scripts/run_engineering_verification_matrix.py --test-id CMP-L2-TRANSMITTER-COMBINED-DEFAULT-SET --output reports//CMP-L2-TRANSMITTER-COMBINED-DEFAULT-SET.json	{"duration_s":600,"finite_numeric_required":true}	At least one fault and one degradation participate in one run.
CMP-BND-TRANSMITTER-SHORT-DURATION-60S	component	transmitter	duration_boundary / short_duration_60s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-TRANSMITTER-SHORT-DURATION-60S --output reports//CMP-BND-TRANSMITTER-SHORT-DURATION-60S.json	{"duration_s":60.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-BND-TRANSMITTER-LONG-DURATION-86400S	component	transmitter	duration_boundary / long_duration_86400s	python scripts/run_engineering_verification_matrix.py --test-id CMP-BND-TRANSMITTER-LONG-DURATION-86400S --output reports//CMP-BND-TRANSMITTER-LONG-DURATION-86400S.json	{"duration_s":86400.0,"finite_numeric_required":true}	Boundary-duration combined sparse event run completes with finite outputs and bounded sample count.
CMP-NEG-TRANSMITTER-	component	transmitter	negative /	python	{"expected_exception":"ValueError"	Unsupported working scenario is

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
UNSUPPORTED-WORKING-SCENARIO			unsupported_working_scenario	scripts/run_engineering_verification_matrix.py --test-id CMP-NEG-TRANSMITTER-UNSUPPORTED-WORKING-SCENARIO --output reports//CMP-NEG-TRANSMITTER-UNSUPPORTED-WORKING-SCENARIO.json	"}	rejected by the component runner.
CMP-DET-TRANSMITTER-REPEAT-STABILITY	component	transmitter	determinism / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id CMP-DET-TRANSMITTER-REPEAT-STABILITY --output reports//CMP-DET-TRANSMITTER-REPEAT-STABILITY.json	{"repeat_count":2,"semantic_hash_equal":true}	Two same-input runs produce the same semantic hash.
SUB-ADCS-ALL-MODES	subsystem	adcs	all_modes / all_modes	python scripts/run_engineering_verification_matrix.py --test-id SUB-ADCS-ALL-MODES --output reports//SUB-ADCS-ALL-MODES.json	{"finite_numeric_required":true}	Subsystem public runner completes, returns structured outputs and contains finite numeric values.
SUB-ADCS-REPEAT-STABILITY	subsystem	adcs	repeat_stability / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id SUB-ADCS-REPEAT-STABILITY --output reports//SUB-ADCS-REPEAT-STABILITY.json	{"finite_numeric_required":true}	Subsystem public runner completes, returns structured outputs and contains finite numeric values.
SUB-COMM-DATA-ALL-MODES	subsystem	comm_data	all_modes / all_modes	python scripts/run_engineeri	{"finite_numeric_required":true}	Subsystem public runner completes, returns structured outputs and contains finite

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				ng_verification_matrix.py --test-id SUB-COMM-DATA-ALL-MODES --output reports//SUB-COMM-DATA-ALL-MODES.json		numeric values.
SUB-COMM-DATA-REPEAT-STABILITY	subsystem	comm_data	repeat_stability / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id SUB-COMM-DATA-REPEAT-STABILITY --output reports//SUB-COMM-DATA-REPEAT-STABILITY.json	{"finite_numeric_required":true}	Subsystem public runner completes, returns structured outputs and contains finite numeric values.
SUB-EPS-ALL-MODES	subsystem	eps	all_modes / all_modes	python scripts/run_engineering_verification_matrix.py --test-id SUB-EPS-ALL-MODES --output reports//SUB-EPS-ALL-MODES.json	{"finite_numeric_required":true}	Subsystem public runner completes, returns structured outputs and contains finite numeric values.
SUB-EPS-REPEAT-STABILITY	subsystem	eps	repeat_stability / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id SUB-EPS-REPEAT-STABILITY --output reports//SUB-EPS-REPEAT-STABILITY.json	{"finite_numeric_required":true}	Subsystem public runner completes, returns structured outputs and contains finite numeric values.
SUB-PAYLOAD-ALL-MODES	subsystem	payload	all_modes / all_modes	python scripts/run_engineering_verification_matrix.py --test-id SUB-PAYLOAD-ALL-MODES --output reports//SUB-PAYLOAD-ALL-	{"finite_numeric_required":true}	Subsystem public runner completes, returns structured outputs and contains finite numeric values.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				MODES.json		
SUB-PAYLOAD-REPEAT-STABILITY	subsystem	payload	repeat_stability / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id SUB-PAYLOAD-REPEAT-STABILITY --output reports//SUB-PAYLOAD-REPEAT-STABILITY.json	{"finite_numeric_required":true}	Subsystem public runner completes, returns structured outputs and contains finite numeric values.
SUB-PROPULSION-ALL-MODES	subsystem	propulsion	all_modes / all_modes	python scripts/run_engineering_verification_matrix.py --test-id SUB-PROPULSION-ALL-MODES --output reports//SUB-PROPULSION-ALL-MODES.json	{"finite_numeric_required":true}	Subsystem public runner completes, returns structured outputs and contains finite numeric values.
SUB-PROPULSION-REPEAT-STABILITY	subsystem	propulsion	repeat_stability / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id SUB-PROPULSION-REPEAT-STABILITY --output reports//SUB-PROPULSION-REPEAT-STABILITY.json	{"finite_numeric_required":true}	Subsystem public runner completes, returns structured outputs and contains finite numeric values.
SUB-THERMAL-ALL-MODES	subsystem	thermal	all_modes / all_modes	python scripts/run_engineering_verification_matrix.py --test-id SUB-THERMAL-ALL-MODES --output reports//SUB-THERMAL-ALL-MODES.json	{"finite_numeric_required":true}	Subsystem public runner completes, returns structured outputs and contains finite numeric values.
SUB-THERMAL-REPEAT-STABILITY	subsystem	thermal	repeat_stability / repeat_stability	python scripts/run_engineering_verification_matrix.py --test-id SUB-THERMAL-REPEAT-STABILITY --output reports//SUB-THERMAL-REPEAT-STABILITY.json	{"finite_numeric_required":true}	Subsystem public runner completes, returns structured outputs and contains finite numeric values.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				x.py --test-id SUB-THERMAL-REPEAT-STABILITY --output reports//SUB-THERMAL-REPEAT-STABILITY.json		
WHO-REP-001	whole_spacecraft_or_agent_release	component_nominal	representative_run / component_nominal	sat-agent run examples/component_nominal.yaml --output-root runs//WHO-REP-001	{"expected_four_state":true,"input": "examples/component_nominal.yaml"}	Run result matches the documented four-state expectation for the representative case.
WHO-REP-002	whole_spacecraft_or_agent_release	component_fault	representative_run / component_fault	sat-agent run examples/component_fault.yaml --output-root runs//WHO-REP-002	{"expected_four_state":true,"input": "examples/component_fault.yaml"}	Run result matches the documented four-state expectation for the representative case.
WHO-REP-003	whole_spacecraft_or_agent_release	component_degradation	representative_run / component_degradation	sat-agent run examples/component_degradation.yaml --output-root runs//WHO-REP-003	{"expected_four_state":true,"input": "examples/component_degradation.yaml"}	Run result matches the documented four-state expectation for the representative case.
WHO-REP-004	whole_spacecraft_or_agent_release	subsystem_nominal	representative_run / subsystem_nominal	sat-agent run examples/subsystem_nominal.yaml --output-root runs//WHO-REP-004	{"expected_four_state":true,"input": "examples/subsystem_nominal.yaml"}	Run result matches the documented four-state expectation for the representative case.
WHO-REP-005	whole_spacecraft_or_agent_release	whole_spacecraft_combined	representative_run / whole_spacecraft_combined	sat-agent run examples/whole_spacecraft_combined_nominal.yaml --output-root runs//WHO-REP-005	{"expected_four_state":true,"input": "examples/whole_spacecraft_combined_nominal.yaml"}	Run result matches the documented four-state expectation for the representative case.
WHO-REP-006	whole_spacecraft_or_agent_release	mission_fail_preserved	representative_run / mission_fail_preserved	sat-agent run examples/mission_fail_preserved.yaml --output-root runs//WHO-REP-006	{"expected_four_state":true,"input": "examples/mission_fail_preserved.yaml"}	Run result matches the documented four-state expectation for the representative case.
ENV-001	environment	environment	version / version	sat-agent version	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
ENV-002	environment	environment	strict_doctor / strict_doctor	sat-agent doctor --require-api --strict-assets	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
STA-001	static_quality	static_quality	compileall / compileall	python -m compileall -q src scripts tests	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
STA-002	static_quality	static_quality	ruff_blockers / ruff_blockers	ruff check src tests scripts --select F821,F822,F811,F841	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
STA-003	static_quality	static_quality	duplicate_definitions / duplicate_definitions	python scripts/check_no_duplicate_module_definitions.py	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
STA-004	static_quality	static_quality	public_exports / public_exports	python scripts/check_public_module_exports.py	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
STA-005	static_quality	static_quality	public_type_hints / public_type_hints	python scripts/check_public_type_hints.py	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
STA-006	static_quality	static_quality	exception_policy / exception_policy	python scripts/check_exception_policy_既有工程版本.py	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
CAL-001	calibration	calibration	readiness_check / readiness_check	sat-agent calibration check --project-root .	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
CAL-002	calibration	calibration	readiness_report / readiness_report	sat-agent calibration report --project-root .	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
CAL-003	calibration	calibration	schema / schema	sat-agent calibration schema --kind all	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
CAL-004	calibration	calibration	invalid_ground_calibrated_rejection / invalid_ground_calibrated_rejection	sat-agent calibration validate examples/calibration/invalid_ground_calibrated_missing_data.yaml --project-root .	{"expected_exit_codes":{1}}	Command exits with expected return code [1] and writes no unhandled exception trace.
MOD-001	model_contract	model_contract	layer_schema / layer_schema	python scripts/check_layer_schema_contract.py	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
MOD-002	model_contract	model_contract	native_capability_audit / native_capability_audit	python scripts/check_native_capability_audit_v2.py	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
MOD-003	model_contract	model_contract	parameter_provenance / parameter_provenance	python scripts/check_parameter_provenance_calibration_1.py	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
MOD-004	model_contract	model_contract	subsystem_builders / subsystem_builders	python scripts/check_subsystem_basisk_builders.py	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
MOD-005	model_contract	model_contract	subsystem_runners / subsystem_runners	python scripts/check_subsystem_basisk_runners.py	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
MOD-006	model_contract	model_contract	runtime_injection / runtime_injection	python scripts/check_runtime_injection_closure_2.py	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
MOD-007	model_contract	model_contract	whole_spacecraft_native / whole_spacecraft_native	python scripts/check_wholesc_native_closure_1.py	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
AGT-001	agent	agent	parse_template / parse_template	sat-agent parse '创建电池部件正常仿真，运行 300 秒' --backend template	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
AGT-002	agent	agent	validate_component / validate_component	sat-agent validate examples/component_nominal.yaml	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
AGT-003	agent	agent	resolve_component / resolve_component	sat-agent resolve examples/component_nominal.yaml	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
AGT-004	agent	agent	plan_whole / plan_whole	sat-agent plan examples/whole_spacecraft_combined_nominal.yaml	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
AGT-005	agent	agent	export_script / export_script	sat-agent export-script examples/component_nominal.yaml	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no

Test ID	层级	对象	场景类型/场景	执行命令	判读参数	通过规则
				ml --output generated/既有工程版本_component_nominal.py		unhandled exception trace.
AGT-006	agent	agent	golden_repeat5 / golden_repeat5	sat-agent eval --repeat-count 5 --execute-marked --output-dir reports/既有工程版本_golden	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
REL-001	release	release	pytest / pytest	python -m pytest -q	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.
REL-002	release	release	release_check / release_check	sat-agent release-check --golden-report reports/agent_golden/golden_eval_report.json --output-dir reports/既有工程版本_release_check --source-root . --strict-assets	{"expected_exit_codes":{0}}	Command exits with expected return code [0] and writes no unhandled exception trace.

7. 失败项处置要求

失败类型	定位文件	处置要求
组件运行失败	reports//<TEST_ID>.json	检查 result_preview、reason、traceback 和组件 runner。
QoI 或数值非有限	reports//<TEST_ID>.json	检查 NaN/Inf 路径，禁止继续声明 PASS。
运行时注入未参与	trace.active_faults / active_degradations	检查事件时间窗、目标参数、default_faults/default_degradations。
Shell 命令失败	stdout_tail/stderr_tail	按返回码和 traceback 定位环境、资产或命令问题。
Golden Set 失败	reports/既有工程版本_golden	检查 TaskSpec hash、Plan hash、Claim hash 和失败案例。

8. 归档要求

正式验收需归档：矩阵 JSON、每项测试 JSON、summary.json、Golden Set 报告、release-check 报告、doctor 报告、制品 SHA-256、Run Bundle 和失败项处置记录。